Ovine/Caprine Infectious Disease

Chi-Fei Kao

2026.10.06

Malignant Catarrhal Fever (MCF)

Malignant Catarrhal Fever (MCF)

- MCF is a frequently fatal, **lymphoproliferative** disease caused by a groups of viruses that belong to the family *Herpesviridae*, subfamily ***Gammaherpesvirinae***, genus ***Macavirus***.
- MCF affects **mainly ruminants** and, rarely, swine.
- European report of sheep-associated MCF in 1798
- In 1850, the Maasai people in Africa discovered that cattle in close contact with wildebeests are prone to MCF.

Etiology

- Two major groups of MCF viruses:
  - ✓**Alcelaphine herpesvirus-1 (AlHV1)**, the wildebeest-associated MCF virus; Alcelaphine MCF occur in most African countries.
  - ✓**Ovine herpesvirus-2 (OvHV2)**, the sheep-associated MCF virus; Sheep-Associated MCF occur worldwide; causes MCF in cervids
- Other MCF viruses:
  - ✓Caprine herpesvirus 2 (CpHV-2), the caprine-associated MCF virus; endemic in most domesticated goat populations worldwide and causes MCF in cervids.

牛羚

鹿科

台灣獸醫誌 Taiwan Vet J 33 (3&4): 162-166, 2007

Short Communication：Malignant Catarrhal Fever Associated with Caprine Herpesvirus-2 Infection in Captive Formosan Sika Deer（*Cervus Nippon Taiouanus*）in Taiwan

Shih-Ping CHEN, Ming-Chung LEE, Yu-Fen SUN, Chi-Min CHEN, Ping-Chen YANG, *Ivan-Chen CHENG

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ABSTRACT Caprine herpesvirus-2（CpHV-2）infection in Formosan sika deer（*Cervus nippon taiouanus*）with clinical signs and pathological changes of chronic malignant catarrhal fever（MCF）detected by polymerase chain reaction（PCR）and sequencing in Taiwan is reported here. During the period of September 2002 and January 2003, 12 Formosan sika deer died with progressive weight loss; while 2 other deer in the same captive herd were subclinically infected. At necropsy, gross lesions were characterized by marked arteriosclerosis in the heart, mesentery, and kidneys were noted. Microscopically, all 12 animals had severe lymphocytic vasculitis with intimal hyperplasia that were consistent with the lesions of chronic MCF. Antibody to MCF virus was detected in sera collected from all deer and goats housing in the same barn. Clinically normal goats, but not cattle, in the same barn were also PCR positive for CpHV-2. Subclinically CpHV-2-infected goats were considered to be the source of infection in this incidence. [Shih-Ping CHEN, Ming-Chung LEE, Yu-Fen SUN, Chi-Min CHEN,Ping-Chen YANG, *Ivan-Chen CHENG. Short Communication：Malignant catarrhal fever associated with caprine herpesvirus-2 infection in captive Formosan sika deer（*Cervus nippon taiouanus*）in Taiwan. Taiwan Vet J 33(3&4): 162-166, 2007. *Corresponding author TEL: (037)585-868, FAX: (037) 585-850, E-mail: icc01@mail.atit.org.tw]

ORIGINAL ARTICLE

Transboundary and Emerging Diseases WILEY

Malignant catarrhal fever: An emerging yet neglected disease in captive sika deer (*Cervus nippon*) herds in China

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Abstract

Malignant catarrhal fever (MCF) is a fatal lymphoproliferative disease that represents a serious problem in the deer-rearing industry. To better understand an MCF-like disease that has emerged in northern China since 2015, we investigated ten cases by documenting clinical and epidemiological data and analysing causative agents and histopathological changes. In addition, a retrospective screen for *Macavirus* DNA and a questionnaire-based survey were conducted. Epizootic MCF in Chinese sika deer herds has emerged with a low morbidity of 3.8% (95% CI: 2.5%–5.1%) and a high mortality of 93.2% (95% CI: 86.6%–99.9%). The disease course varied from 3 to 12 days. Aetiologically, OvHV-2 was predominant in the MCFV, accounting for most MCF cases (21/23). In contrast, only two CpHV-2 isolates were phylogenetically closely related to CpHV-2. Diarrhoea and nasal discharges were the most frequent manifestations, although clinical signs varied in some cases. Pathologically typical lesions of haemorrhage, necrosis and lymphoid cell infiltration were readily observed in a variety of organs. Vasculitis caused by vascular and perivascular lymphoid cell infiltration was common. The retrospective survey suggested a low positive rate (3/275) of MCFV DNA in peripheral blood lymphocytes (PBL). The questionnaire-based survey suggested the disease was neglected by local veterinarians, who did not acknowledge the risk of co-rearing deer with reservoir species. Collectively, the emerging epizootic MCF in Chinese sika deer herds remains neglected, emphasizing the urgency of initiating full-field diagnoses and control strategies.

KEYWORDS
CpHV-2, malignant catarrhal fever, OvHV-2, questionnaire-based survey, retrospective screen, Sika deer

Epidemiology

- **Reservoir** hosts (sheep, goats, wildebeest)
- **Clinically susceptible** (dead-end) hosts (cattle, bison, deer), which **do not shed virus**
- The **morbidity rate varies**; usually sporadic and presents as a single or small number of cases.
- The disease with both agents is almost always fatal in **cattle** and is an important disease of farmed **deer**.

Transmission

- AlHV1

  Transmission in wildebeest occurs mainly perinatally, but some calves may become infected in utero; all calves become infected within the first few months of life and remain carriers for life.

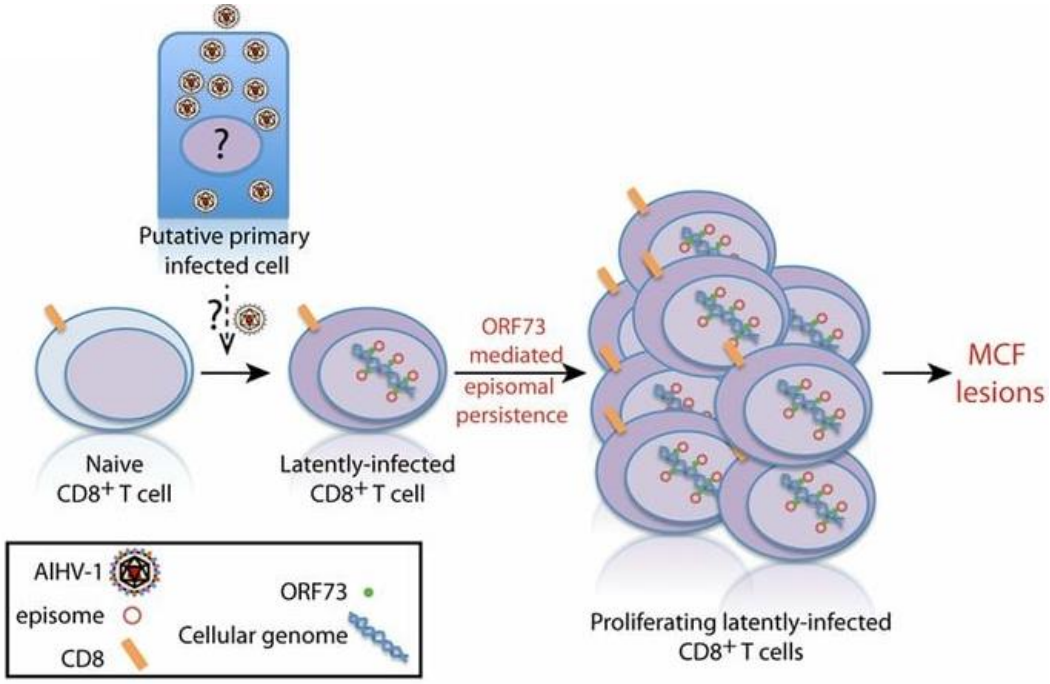
- OvHV2:

  A proportion of lambs are infected in utero with most lambs becoming infected when 3–6 months old by aerosol transmission

- Both AlHV1 and OvHV2 transmitted to susceptible host by **contact** or **aerosol**, primarily from **respiratory secretions** of wildebeest calves (AlHV1) and weaned lambs (OvHV2) under 1 year of age.

Pathogenesis

- Not fully understood
- Vessel-associated lymphoid proliferation
- **CD8+ T-lymphocytes** are the predominant cells associated with the vascular lesions.
- Direct virus–cell interactions vs immune-mediated responses directed against infected cells?



Clinical Presentation

- Incubation period in natural infection varies from 3 to 8 weeks
- Different forms of presentation that are actually **gradations**
  - ✓ Peracute
  - ✓ Alimentary tract form
  - ✓ **Common "head-and-eye" form**
  - ✓ Mild form

→All of these forms may be produced during infection with one strain.

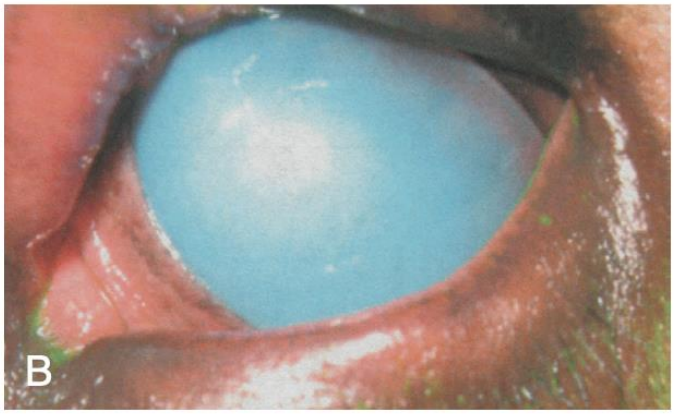
- In cattle, lymphadenopathy and severe eye lesions are more frequent, and hemorrhagic enteritis and cystitis less frequent, than in deer and bison.
- The presence of nasal discharge, corneal edema, fever, and lymphadenopathy is very helpful in differentiating MCF from mucosal disease or bluetongue.

Head-and-Eye Form

- Extreme depression & anorexia
- Agalactia
- High fever (41° to 41.5° C)
- Profuse mucopurulent nasal discharge
- Severe dyspnea with stertor
- Ocular discharge with variable degrees of corneal edema
- Blepharospasm (眼瞼痙攣) and uveitis
- Congestion of scleral vessels
- **Superficial necrosis** is evident in the anterior nasal mucosa and on the buccal mucosa







https://www.askjpc.org/wsco/do_slideshow2.php?pic_series=bHdiQUkvNGhCUUVpeVBhRUd0bVBaZz09&cYear=2022&initImage=cE9lVU5sSjJJdHlEd2x0OGdLdXdqQT09



Malignant Cattarrhal Fever (MCF). in Western part of Iran.

https://www.youtube.com/watch?v=Oawgx1b8fB4

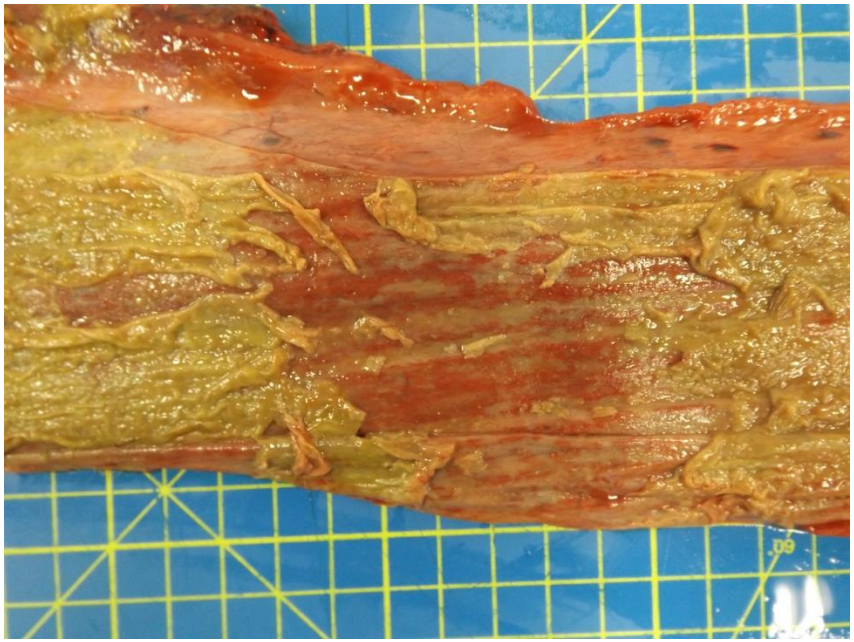
In chronic case skin lesions including local papule formation and eczematous (濕疹樣) weeping occur.



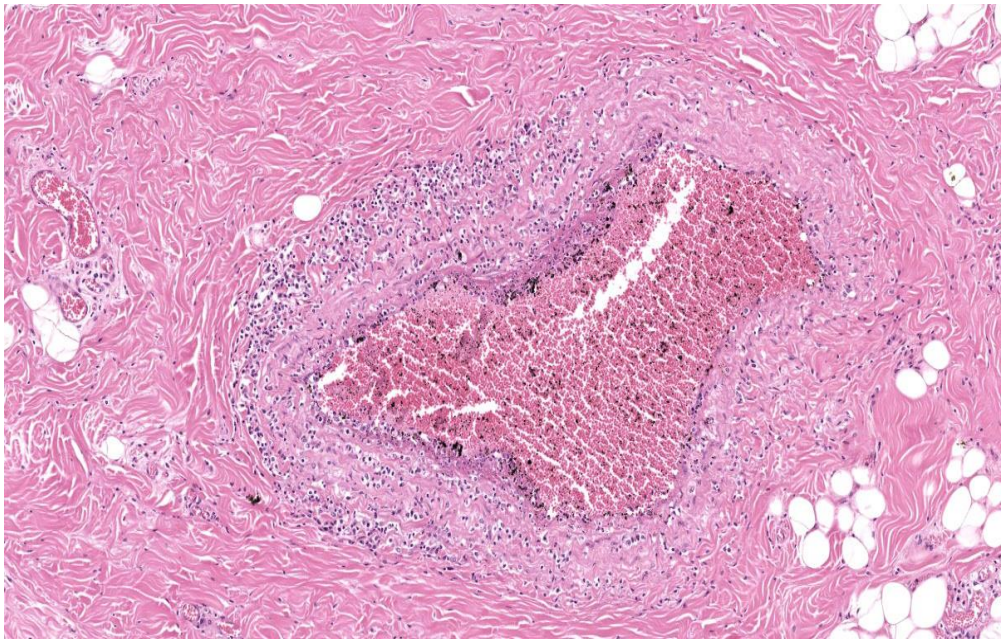


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Pathological Findings



Lesions in the mouth, nasal cavities, and pharynx vary from minor degrees of hemorrhage and erythema through extensive, severe inflammation to discrete ulcers.



The pathognomonic lesion is a necrotizing vasculitis that features infiltration of the tunica media and adventitia by lymphoblast-like cells and macrophages.

https://www.askjpc.org/wsco/do_slideshow2.php?pic_series=bHdiQUkvNGhCUUVpeVBhRUd0bVBaZz09&cYear=2022&initImage=cE9lVU5sSjJJdHlEd2x0OGdLdXdqQT09

Prevention & Control

- No effective vaccine is currently available.
- Sheep can be produced that are free of virus by early weaning and isolation.
- Separation of carriers from susceptible species.

Bluetongue

Introduction

Culicoides biting midges庫蠓(糠蚊)

- An **arthropod-borne**, noncontagious viral disease primarily of domestic and wild ruminants (eg. white-tailed deer).
- Cause by bluetongue virus (BTV), an **Orbivirus** in the family *Reoviridae* with a genome composed of 10 dsRNA segments.

  → **several (at least 29) serotypes** and considerable genetic heterogeneity.
- The occurrence of different BTV serotypes varies by geographic region.

  → the occurrence and activity of the *Culicoides* vectors
- **WOAH notifiable disease**.

「藍舌病」在歐洲蔓延 畜牧業牛羊加快施打疫苗

#施打疫苗 #疫苗接種 #農場 #病毒

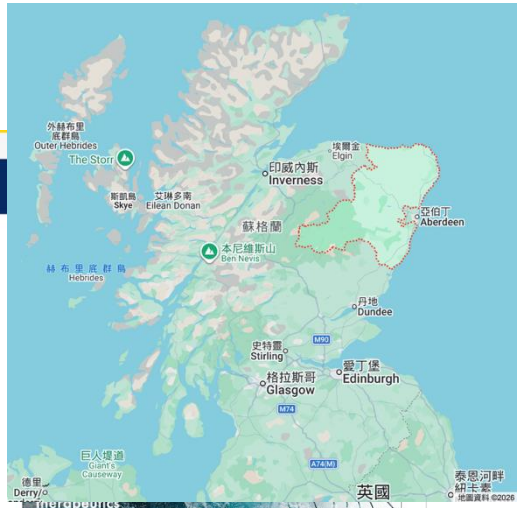


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梁雁 / 編譯

結論先講

近日歐洲多個國家傳出藍舌病迅速蔓延，這種病毒會造成牛、羊出現高燒和舌頭發藍等症狀，最嚴重可能會致死，目前畜牧業者加快施打疫苗的速度。不過，接種過疫苗的牲口，還是可能死於藍舌病，專家和業者不敢掉以輕心。



World US politics UK **Climate crisis** Middle East Ukraine Environment Science Global development Football Tech Business Obituaries

Farming

Worst ever outbreak of bluetongue hits British farms

Climate crisis blamed as virus that originated in Africa spreads to livestock as far north as Aberdeenshire



Pippa Neill *Environment reporter*

Sat 3 Oct 2026 06.00 BST

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"The virus, which is transmitted via biting midges and originated in Africa, has spread to Europe over the past two decades, a development the World Organisation for Animal Health links to climate breakdown. The particular strain of bluetongue currently affecting livestock, serotype 3 (BTV-3), was first found in England in 2023."

Epidemiology

- **Cattle are the reservoir and amplification hosts**.
- **Severe disease** is most common in European fine wool and mutton breeds of **sheep**.
- Certain serotypes (BTV-8) can cause severe disease in cattle.
- Virus concentrations in secretions and excretions are minimal, making direct, indirect, or aerosol transmission unlikely.

  → BTV is transmitted biologically almost exclusively by specific species of *Culicoides*庫蠓屬.

Epidemiology

- **Transplacental infection** (sheep and cattle)
  - ✓Infection with modified live attenuated or vaccine strains
  - ✓Serotype 8
- Outcome of transplacental infection depends on the stage of pregnancy: from **abortion** to congenital defects.
- BTV infection in cattle is considered to be transient and **neither** persistent **nor** immunotolerant.

- Venereal transmission has been documented but is not considered a significant mechanism of transmission.

Eg. the re-emergence of BTV-8 in France in 2015

國立臺灣大學獸醫專業學院獸醫學研究所

博士論文

Graduate Institute of Veterinary Medicine

School of Veterinary Medicine

National Taiwan University

Doctoral Dissertation

臺灣藍舌病病毒感染及病毒株性狀分析

Bluetongue Virus Infection in Taiwan and Characterization of Taiwanese Virus Strains

李璠

Fan Lee

指導教授：王汎熒博士

Advisor: Professor Fun-In Wang, D.V.M., Ph.D

中華民國100年6月

June, 2011

中文摘要

Previous study demonstrate low virulence

藍舌病主要感染畜養及野生的反芻動物，為家畜最重要的動物傳染病之一。我國在 2003 年曾分離到二株藍舌病病毒，分別命名為 BTV2/KM/2003 與 BTV12/PT/2003。本論文紀錄了對我國藍舌病病毒感染近況的調查以及回溯性研究，並運用綿羊人工感染試驗、病毒核酸序列分析、親緣關係分析、病毒基因轉錄之觀察，對國內的藍舌病病毒株進行深一層的了解。全國性的血清學調查顯示藍舌病病毒感染在國內的牛及山羊族群相當普遍，最早可溯自 1989 年採集的山羊血清檢出抗體陽性。以 BTV2/KM/2003 病毒株人工接種於無抗體之柯麗黛綿羊驗並未出現任何臨床症狀及病理變化，顯示該病毒株為弱毒株。前述二個病毒株的基因體已完成核苷酸定序，同時亦推演為各基因產物的胺基酸序列。分析其核苷酸序列，發現此二病毒株除了 VP2 基因之外極為相似，親緣關係分析亦顯示二者與大陸、印度、印度尼西亞及日本的藍舌病病毒株較為相近。此外，亦利用即時反轉錄聚合酶連鎖反應檢測感染之幼齡倉鼠細胞株及感染綿羊包埋組織內的 mRNA 含量。在受 BTV2/KM/2003 病毒株感染的細胞中，NS2、VP4、VP7 三種基因的轉錄量最高；然而，在感染綿羊的組織內，三種非結構蛋白基因 NS1、NS2、NS3 的轉錄量最高。這些研究顯示本土藍舌病病毒分離株的弱毒力與田間未出現任何藍舌病臨床病例相互吻合，而國內存在的病毒株屬於亞洲族系。

關鍵詞：藍舌病、血清學盛行率、病毒毒力、親緣關係分析、演化、基因轉錄

Table 3. Seroprevalence of bluetongue virus infection in cattle in Taiwan (2003)

Counties	Cattle (herds)	Cattle (heads)	
	Positive/Tested	Positive/Tested	Prevalence
Taoyuan	3/3	19/90	21.1
Hsinchu	1/2	42/58	72.4
Miaoli	1/1	8/8	100.0
Nantou	1/1	6/47	12.8
Yunlin	13/16	79/268	29.5
Chiayi	4/4	39/135	28.9
Tainan	8/8	182/626	29.1
Kaohsiung	2/2	19/120	15.8
Pingtung	22/25	298/1293	23.0
Yilan	1/1	1/7	14.3
Hualien	3/3	78/118	66.1
Taitung	3/3	94/95	98.9
Kinmen	6/6	109/112	97.3
Total	68/75	974/2977	32.7

Table 4. Seroprevalence of bluetongue virus infection in goats in Taiwan (2003)

Counties	Goat (flocks)	Goat (heads)	
	Positive/Tested	Positive/Tested	Prevalence
Miaoli	1/1	2/30	6.7
Nantou	0/1	0/55	0.0
Yunlin	3/10	7/216	3.2
Chiayi	0/2	0/63	0.0
Tainan	0/3	0/194	0.0
Kaohsiung	0/3	0/118	0.0
Pingtung	3/15	9/450	2.0
Yilan	0/4	0/78	0.0
Taitung	1/1	1/30	3.3
Kinmen	5/5	92/116	79.3
Total	13/45	111/1350	8.2

Morbidity & Mortality

- When the disease occurs in a flock for the first time, the incidence of clinical disease may reach 50% to 75% and the mortality 20% to 50%

Pathogenesis

- There are differences in virulence between serotypes.
- In cattle and wild ruminants by most virus strains, endothelial cell damage is minimal.
- Sporadic clinical cases observed in cattle were thought to be the result of **type I hypersensitivity** reaction triggered by repeated exposure to virus-specific IgE.

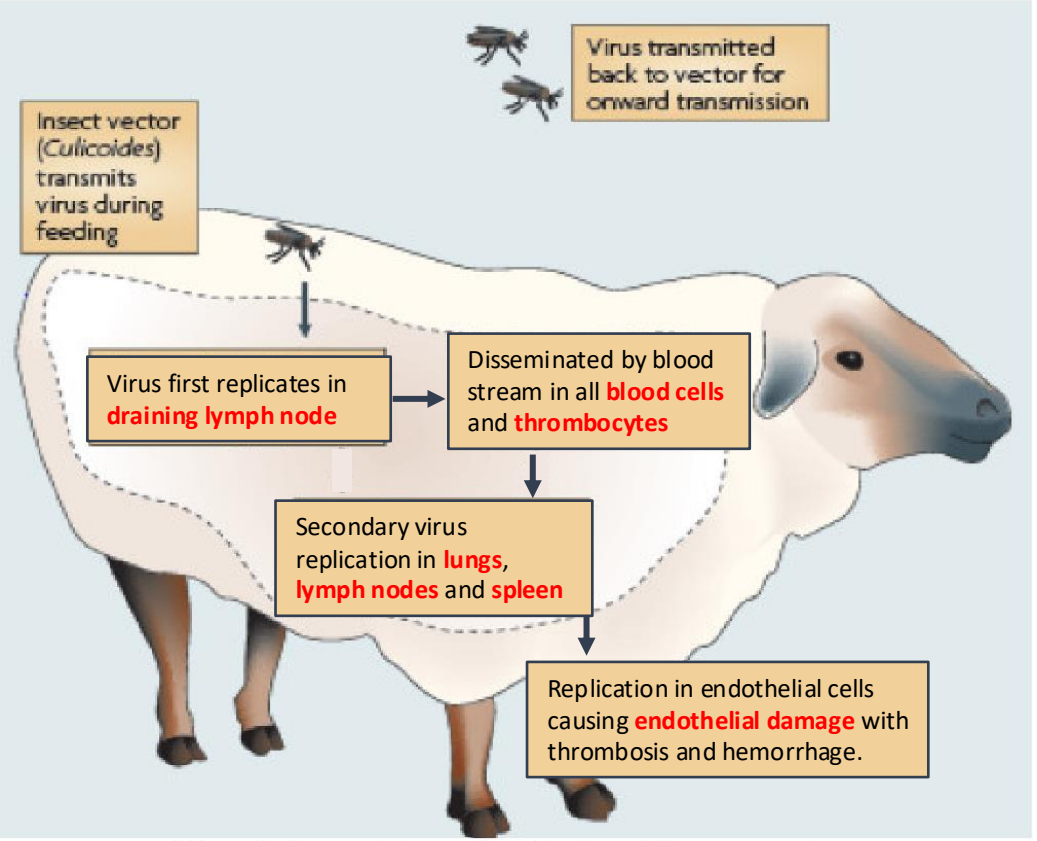


Figure 3; Transmission and replication of bluetongue virus.

Modified from Jilo, 2016

Clinical Presentation

- Sheep (incubation usually < 7days)
  - ✓ **High fever** for 5–6 days
  - ✓ Mucopurulent and often blood-tinged **nasal discharge**.
  - ✓ Salivation
  - ✓ Swelling, reddening and **excoriation of buccal mucosa**
  - ✓ Stertor (respiratory obstruction)
  - ✓ Death in peracute cases occurs about 6 days after the appearance of signs due to asphyxiation.
  - ✓ Some animals recover but there is a **long convalescence**, and a return to normal may take several months
- **Cattle: rare**. Photodermatitis may develop at later stages of the disease (2–3 weeks after infection). Contraction of the infection during early pregnancy may cause abortion, stillbirth, or congenital deformities.

However, in many areas of the world, BTV infection in sheep is subclinical.

Pathological Findings

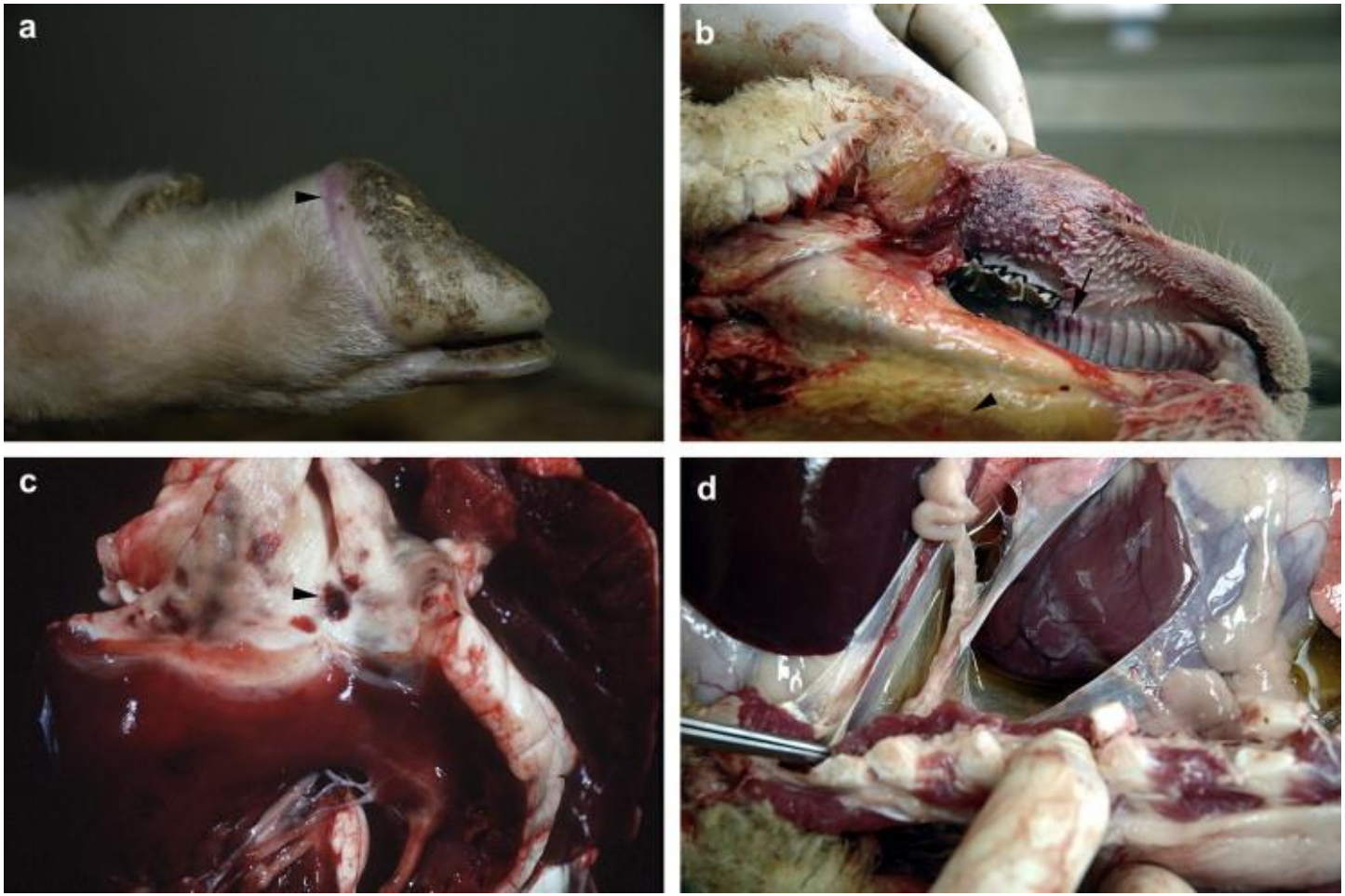
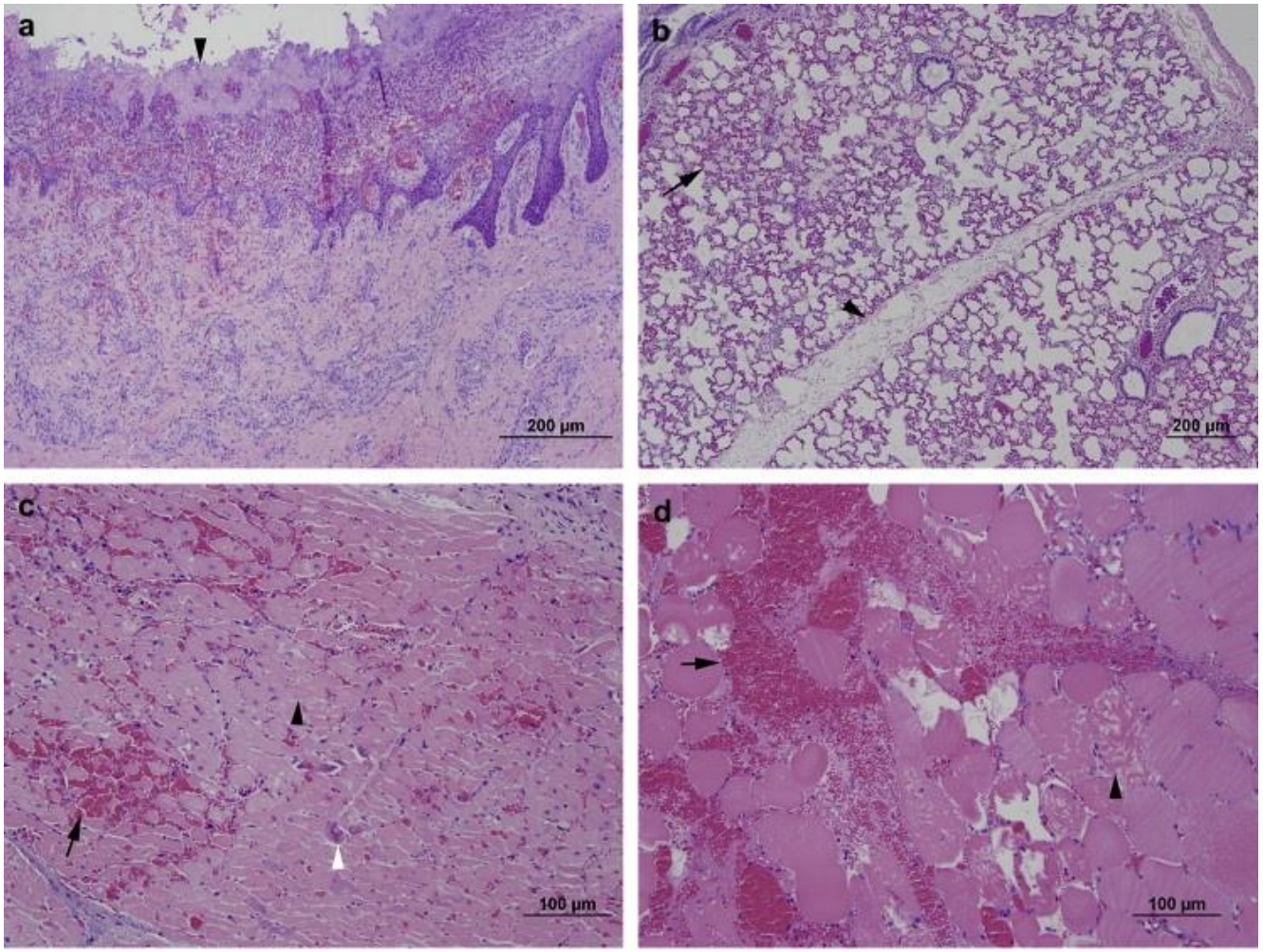


Fig. 2. Gross lesions in sheep with bluetongue. (a) Distal limb: coronitis (arrowhead). (b) Head: submandibular oedema (arrowhead) and haemorrhage and ulceration (arrow) of the hard palate and dental pad. (c) Heart: subintimal haemorrhages (arrowhead) in the pulmonary artery. (d) Pericardial space: pericardial effusion.

Maclachlan et al., 2009



Histopathological lesions in sheep with bluetongue. (a) Buccal mucosa: **superficial epithelial ulceration/erosion** (arrowhead) with submucosal hemorrhage and inflammation. (b) Lung: pulmonary **oedema**, with expansion of the interlobular septae (arrowhead) and accumulation of protein rich fluid in airspaces (arrow). (c) Heart: acute **myocardial necrosis** (arrowhead), **haemorrhage** (arrow) and mineralization (white arrowhead). (d) Skeletal muscle: acute **myonecrosis** (arrowhead) and haemorrhage (arrow).

Maclachlan et al., 2009



12.16. Bluetongue: mucopurulent nasal discharge, gum erosions (Netherlands)



A Dorset Poll sheep with a severely swollen tongue, oedemic face and coronitis. Image: IAH, Pirbright.

Prevention & Control

- **Vector abatement**
- Vaccination is the only satisfactory control procedure once the disease has been introduced into an area.
- Vaccination will **not prevent** or eliminate infection, but it is successful in keeping losses to a very low level.
- **Live attenuated vaccines should not be used in pregnant ewes**.

Sheeppox & Goatpox

Introduction

- **Highly contagious** disease produced by sheeppox (SPPV) and goatpox (GTPV) viruses (both are of the genus *Capripoxvirus*), collectively called capripox infections.
- **WOAH notifiable disease**
- The viruses are **closely related genetically**, and hence they cross-react in serologic tests, and many can cross species barriers in experimental infections.
- Economic loss is from mortality, abortions, mastitis, loss of wool, skin condemnation, and loss of exports


我們做到了!
STOP
守護臺灣羊!
成功撲滅羊痘獲WOAH認同
世界動物衛生組織(WOAH)於112年12月19日認同我國爲羊痘非疫國的聲明，
我國對於動物傳染病防疫的努力，再獲國際組織的認同！
10年以上
無羊痘案例!
未來持續輔導產業升級
生產端
持續補助省工、降溫及自動化設備，降低人工、散熱及搬運等飼養成本，提升飼養效率
屠宰端
補助改善羊隻屠宰設備，減少屠宰損耗，並於屠宰場推動國產羊肉溯源制度，落實產品溯源管理
產品端
持續輔導設置屠宰冷鏈物流設備，以及儲乳槽等儲運設施，確保羊肉及羊乳品質與衛生安全
銷售端
輔導強化國產鮮羊乳GGM標章及國產羊肉專賣店標章認證制度，加強國產羊肉及羊乳廣宣與促銷強化市場區隔

Pathogenesis

- Occurs in all ages, but most severe in young lambs
- The virus is present in **nasal and oral secretions** for several weeks after infection and can live in **scabs** that have fallen off the animal for several months.
- **Transmitted via aerosol or direct contact** with contaminated materials and through skin abrasions produced iatrogenically or by insects.
- High morbidity (endemic areas 70–90%) and variable mortality (endemic areas 5–10%, but can approach 100% (in imported or young animals)

Native breeds in endemic areas are far less susceptible 久了就無感了

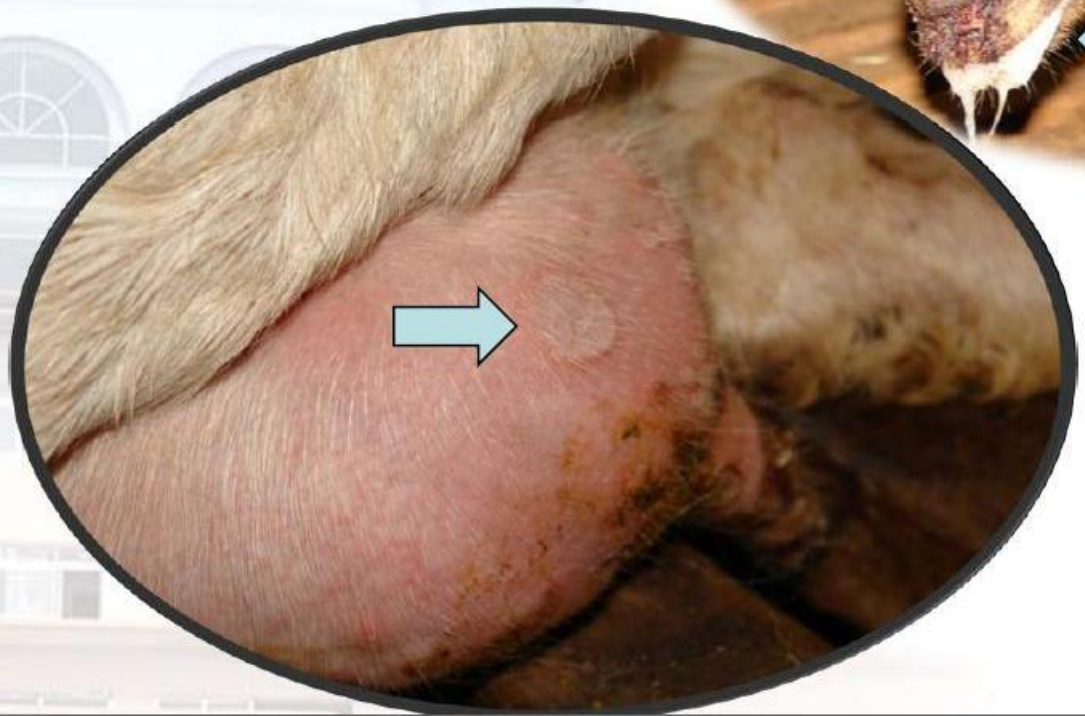
Clinical Presentation

- Similar presentation in both sheeppox & goatpox
  - ✓1–2 weeks of incubation period
  - ✓Fever
  - ✓Generalized lymphadenopathy
  - ✓Swollen eyelids
  - ✓Mucopurulent nasal and lacrimal discharge
  - ✓Widespread skin lesions develop
  - ✓**Malignant form: systemic disease in lambs or kids,** with lesions spread generally over the skin and on the **respiratory and alimentary mucosae.**

山羊痘臨床症狀及肉眼病變

山羊痘發病動物常呈發熱、呼吸困難、眼鼻分泌物多及膿稠嚴重者眼瞼黏連致眼睛無法張開、乳房亦可見皮膚痘瘡、潰瘍及乳房炎等病變。**(傳染性化膿性皮膚炎不會有前述病變)**





山羊痘臨床症狀及肉眼病變

山羊痘對動物之致病性高，全齡羊隻皆會感染，傳播快速，除造成嚴重之全身皮膚炎病變外，亦會導致肺水腫及肺炎，另其於粘膜面也會造成痘狀病變，常會繼發二次感染導致動物有高發病率及死亡率。發病動物常呈發熱、呼吸困難、乳房炎、皮膚痘瘡及潰瘍等病變，懷孕母畜也會有流產現象。

特別是兩側腹部皮膚及尾根部皮膚會有丘疹樣結節，若發現臉口唇部有丘疹或痘瘡病灶，一定要觸診腹部兩側，檢查是否有隆起之丘疹或痘瘡(因毛髮之覆蓋若不觸診或將毛髮剃除容易將此特殊病灶遺漏)



將毛髮剃除後發現明顯隆起丘疹病灶

臉口唇部有痘瘡病灶

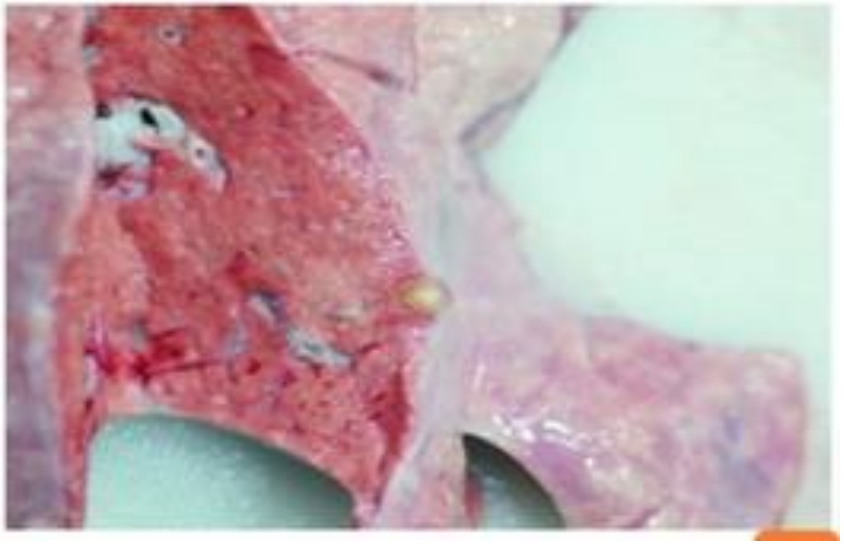


Pathological Findings

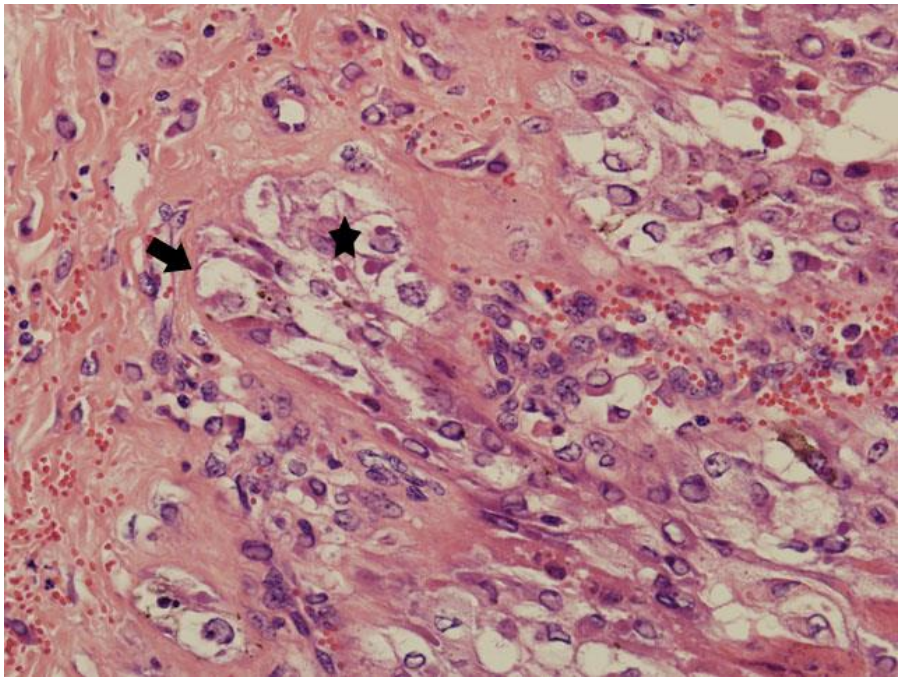




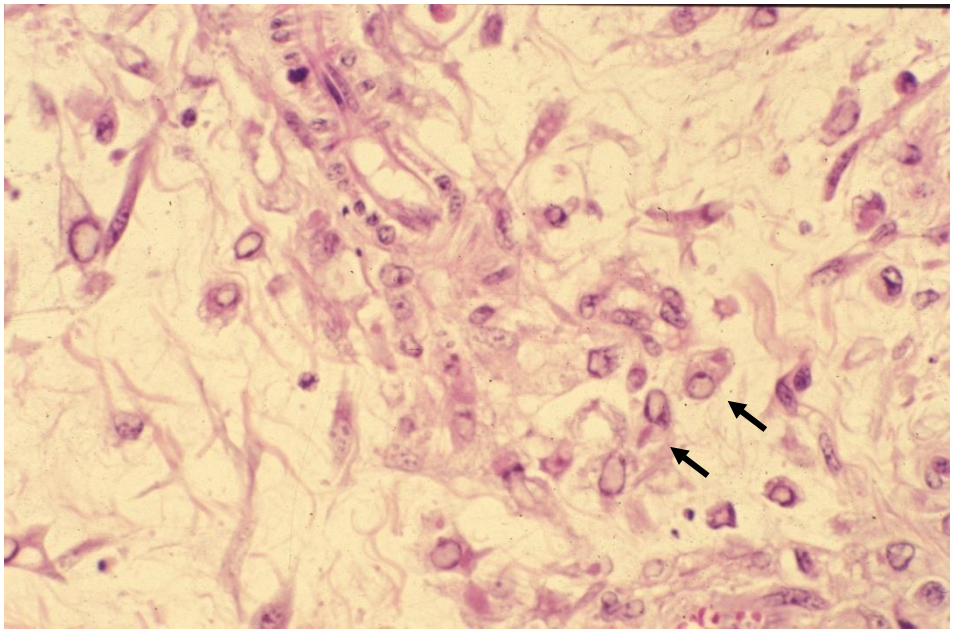




https://www.angrin.tlri.gov.tw/goat/gfa55/gfa55p22-31.htm



Within the stratum basale there is **hydropic degeneration** characterized by abundant intracellular edema (arrow) and low numbers of keratinocytes contain intracytoplasmic, 5-7 micron, **eosinophilic intracytoplasmic inclusion bodies** (star)



Extensive dermal edema during the papular phase with accumulation of many mononuclear cells with eosinophilic ICIB (arrow) around blood vessels and between collagen bundles.

***Sheeppox cells** are virus-infected monocytes, macrophages, and fibroblasts, but **not** endothelial or epithelial cells.

https://www.askjpc.org/wsco/do_slideshow2.php?pic_series=ZTgxazVYbUlYUFBsbkhOK0YxelB6UT09&cYear=2008&initImage=cDRTRUNLNmw0NTIwRUs1ZkVYdStQdz09



COVID-19 疫苗
請參閱衛生福利部疾病管制署提供的最新資訊.
瞭解詳情
在 Google 上查看更多相關資訊

臺灣羊痘事件紀實【防檢署APHIA】

https://www.youtube.com/watch?v=sN5802ysd1I

Prevention & Control

- Prohibiting the importation of live animals and unprocessed produce from infected areas.
- **Ring vaccination and the destruction** of affected flocks if introduced.
- Vaccination with a single capripox vaccine will give **protection across all species and against all capripox infections.**

Sheeppotx, gaotpox, lumpy skin disease

- Killed virus vaccines elicit only temporary protection, but live attenuated vaccines protect against infection for more than 1 year.

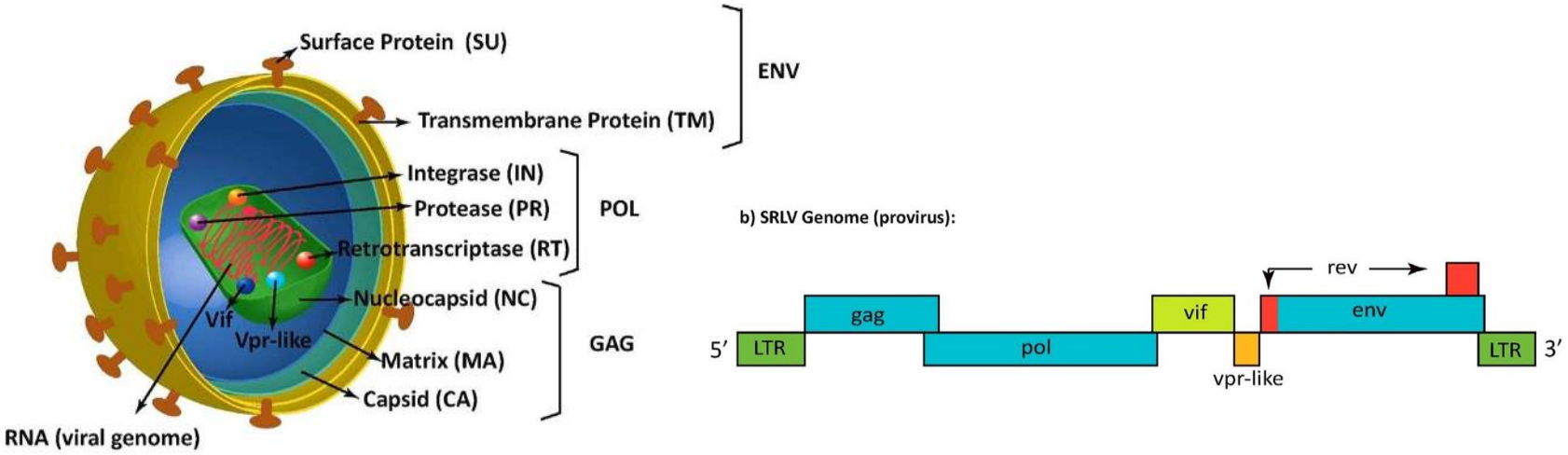
Caprine Arthritis Encephalitis (CAE)

Introduction

- A **persistent lentiviral infection** of goats
- Multiple clinical presentations:
  - ✓Leukoencephalomyelitis, affecting young kids
  - ✓Chronic, hyperplastic polysynovitis
  - ✓Indurative mastitis
  - ✓Interstitial pneumonia.
- **WOAH notifiable disease**
- First described in 1974. CAE is now an important infectious disease of goats **worldwide**. In some herds, the infection rate is as high as 80%.
- Taiwan first confirmed the presence of CAE through pathology and EM in 1993.

Etiology

- Caprine arthritis encephalitis virus, CAEV
- Genus *Lentivirus, Orthoretrovirinae*, family of *Retroviridae*
- It is often grouped together with Maedi-visna (MV) of sheeps, also known as ovine progressive pneumonia (OPP), as the **small ruminant lentiviruses (SRLVs)**



Epidemiology

- The overall seropositive rate was 61.7% (2,120/3,437) in goats and in 98.5% (64/65) of goat farms. These results provide the first large-scale serological evidence for the presence of CAEV infection, indicating that the disease is **widespread in Taiwan**.

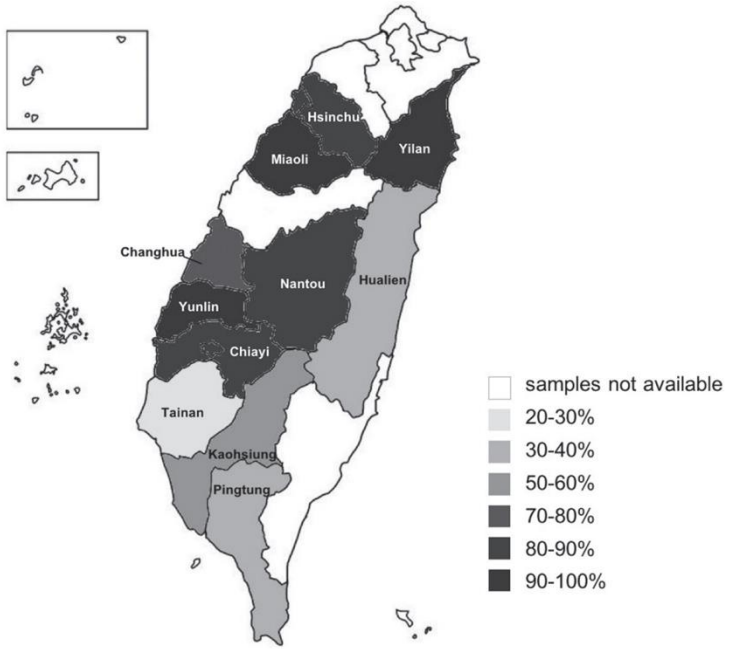


Fig. 1. Geographic distribution of seropositive goats for anti-CAEV antibodies in Taiwan, 2011–2012 (n=3,437). Almost all the goats from each county in Taiwan we tested were seropositive for anti-CAEV antibodies. The seropositive rates ranged from 49.7 to 90.7% among four regions of Taiwan.

Table 2. Prevalence of CAEV infection status by the types of goat farms

Types of farms	n[a)]	Prevalence of CAEV[b)]			Chi-square	*P* value
		+	−	+/−		
Dairy goat	2,035	1,669 (82.0)	339 (16.7)	27 (1.3)	932.98	<0.0001
Meat goat	1,138	399 (35.1)	731 (64.2)	8 (0.7)		
Breeding goat	264	52 (19.7)	212 (80.3)	0 (0.0)		

a) n: sample size, the percentage in parenthesis. b) CAEV ELISA results: +: positive, −: negative, +/−: suspected positive.

Yeang et al., 2017

Transmission

- All breeds and all ages are susceptible to infection; differences, if occur, likely reflect differences in management
- **Ingestion of infected colostrum or milk**.
- The presence of antibody in colostrum does not prevent infection.

  → Neonates should be removed from the doe immediately after birth and reared on pasteurized milk, and many can be reared free from infection
- **Contact transmission** will result in the spread of the disease.
- *In utero* (rare)

Pathogenesis

- Animals infected at birth remain **persistently infected** for life, although only a proportion, typically from 10% to 30%, will develop clinical disease.
- The virus **infects a variety of cell types** including choroid plexus, mammary epithelium, fibroblasts, endothelial cells, and monocytes; however, **replication primarily takes place in mature macrophages**; the virus is activated when monocytes mature into macrophages and is **persistent in macrophages**, **microglia, monocytes, and dendritic cells**
- Lesions (lymphoproliferative) have been shown to be at least partially **immune-mediated** as well as due to direct destructive effect of virus

Clinical Presentation

- Most infected goats are **asymptomatic** (~70%)
- Neurologic syndrome (2-4 months old):
  - ✓Neurologic signs are due to motor spinal dysfunction without signs of cerebral disease
  - ✓Clinical signs are progressive: Caudal ataxia and trembling of the lips → Afebrile; progressive posterior ataxia and extensor paralysis and tetraplegia → death from starvation (wasting) or secondary infection
  - ✓Once paralytic signs develop, death is certain
- Arthritis: Lameness, joint effusion, enlarged carpal joints, high prevalence of carpal hygromas
- Interstitial pneumonia: Dyspnea and coughing, cachexia
- Mastitis: Udder firm, swollen; reduced milk production

Leukoencephalomyelitis





Arthritis

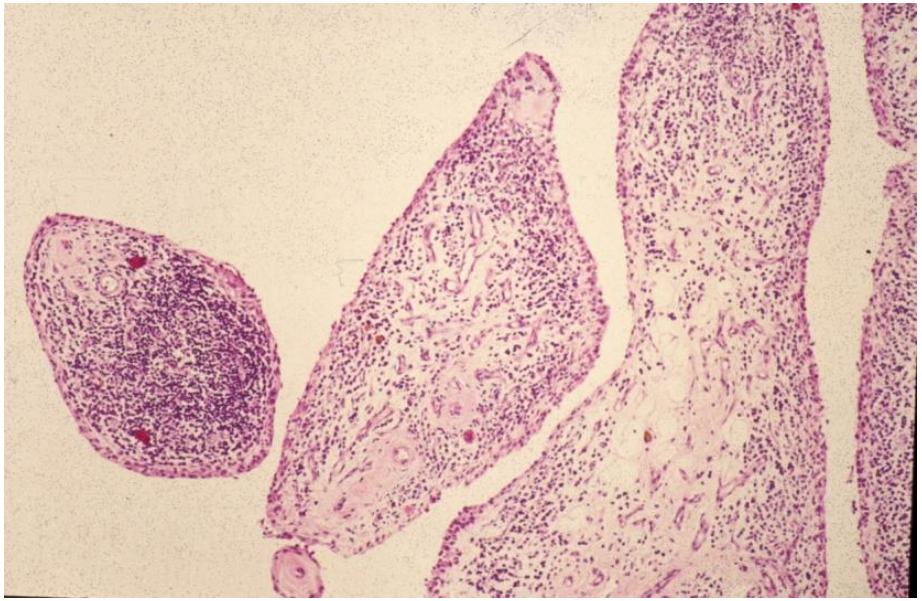




Pathological Findings–Arthritis



Chronic proliferative arthritis/synovitis of carpal, stifle, hock joints; carpal hygromas



Marked villous proliferative synovitis with massive lymphoplasmacytic and histiocytic infiltration.

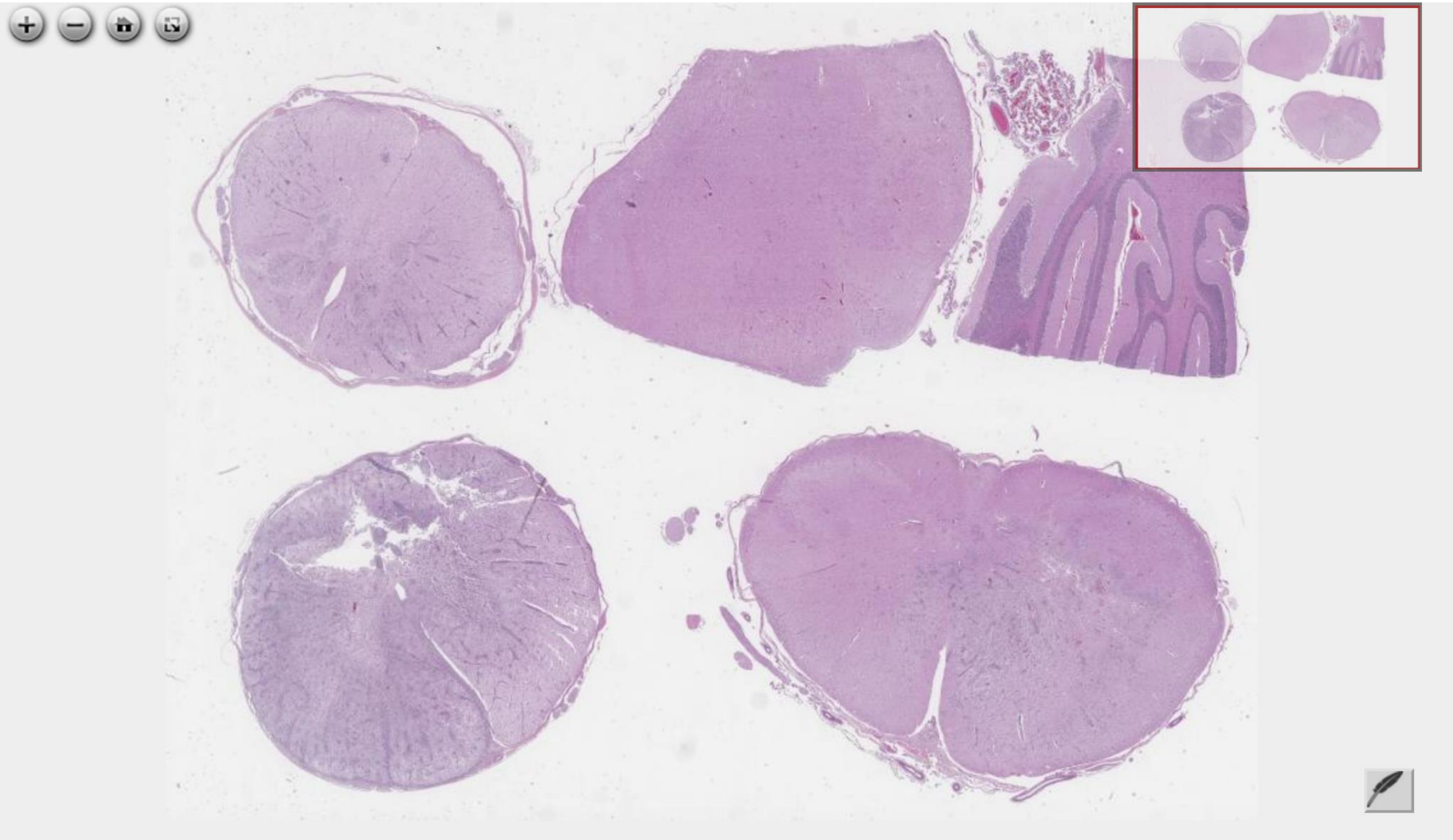
Pathological Findings–CNS (case)

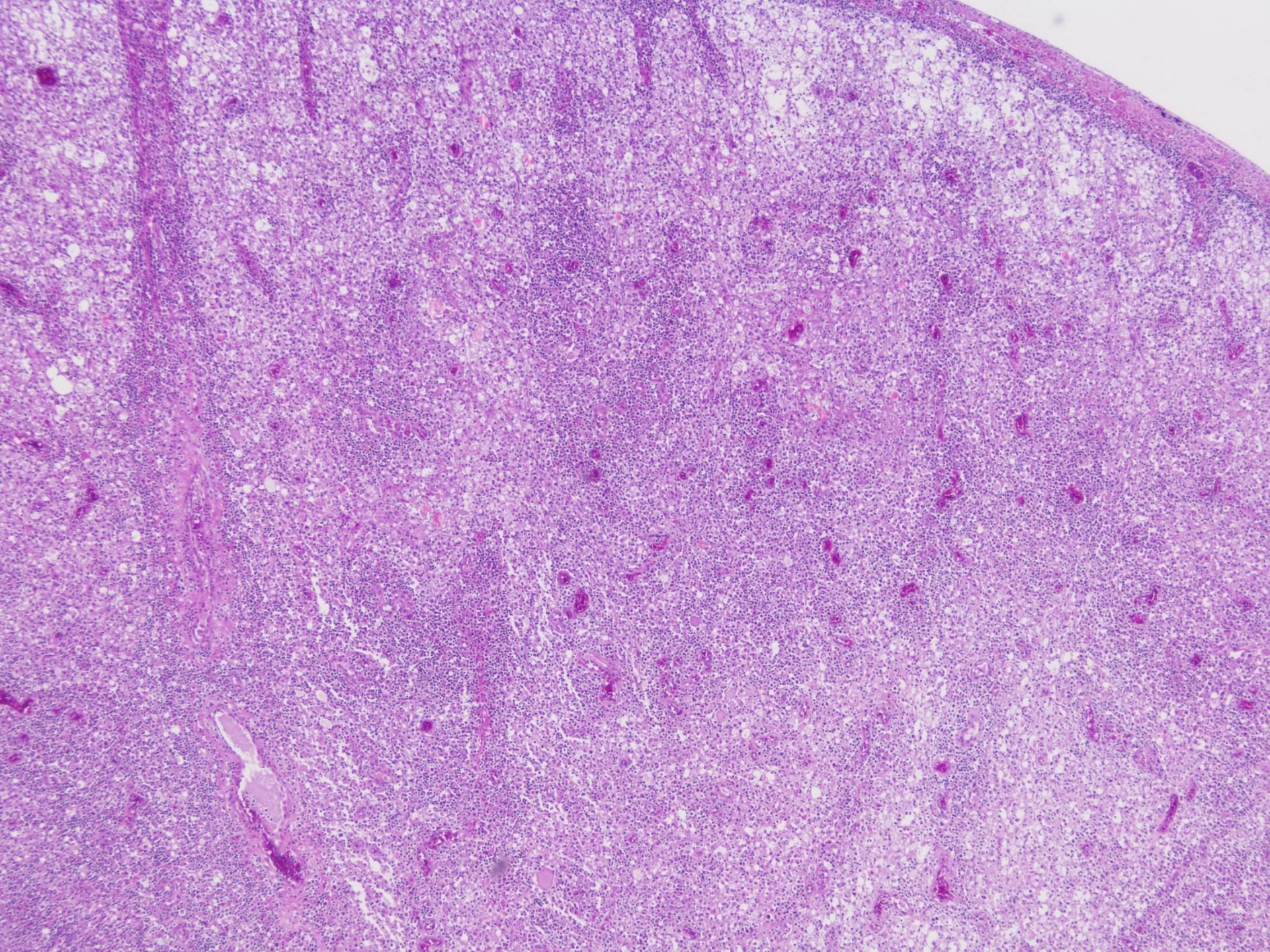
- Three-month old, intact female, Alpine goat
- Respiratory signs were noted for few months.
- The patient was admitted to NTUVH on 2016.11.24 due to paralysis of left forelimb and bilateral hindlimbs. The spirit of the patient remained normal at that time.
- Right-beating nystagmus developed after hospitalization.
- Neurological examination indicated the lesions were located at L2 and forebrain.
- The patient was euthanized.

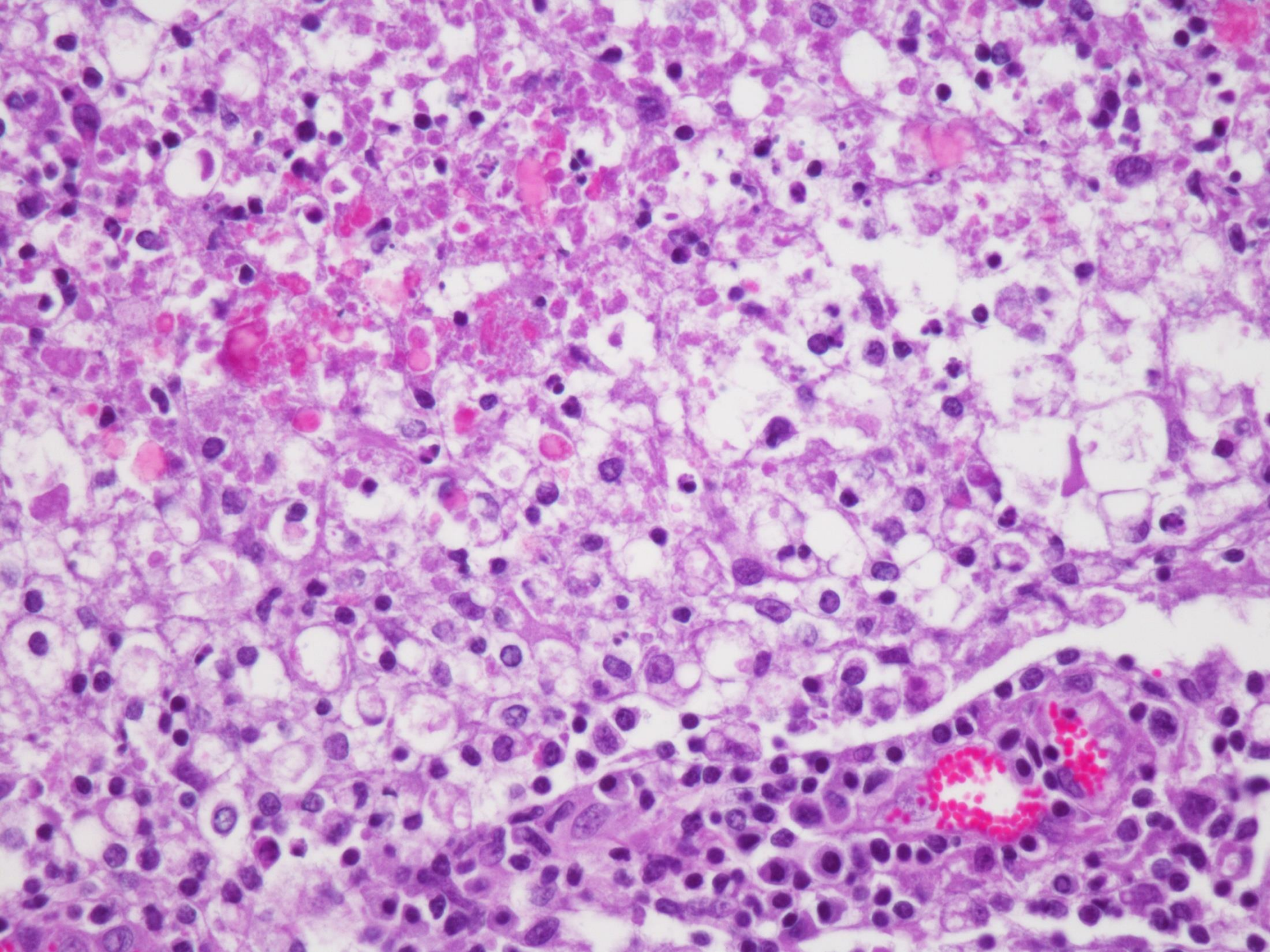






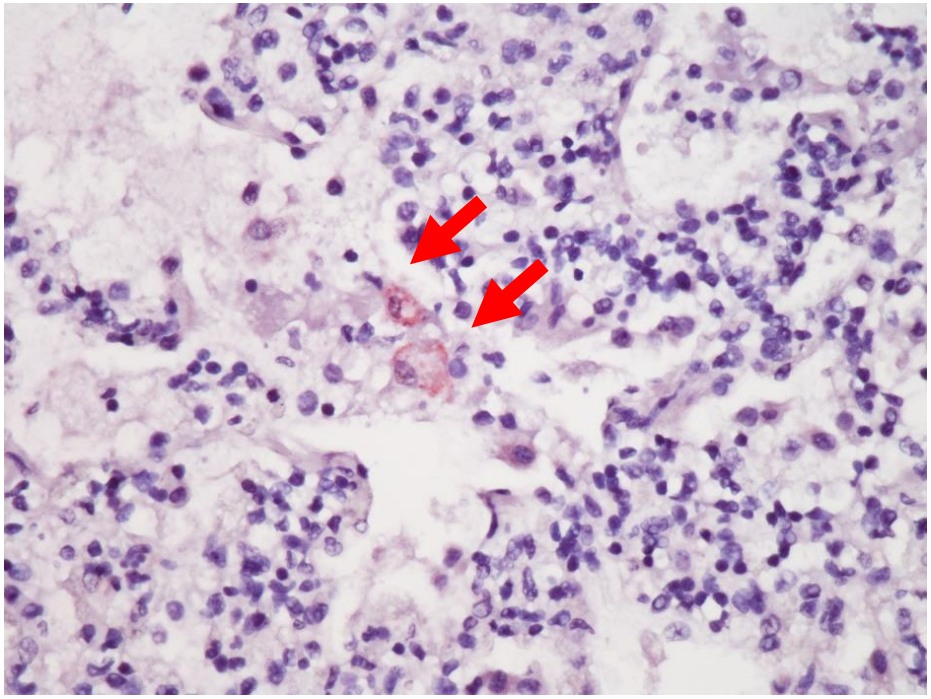


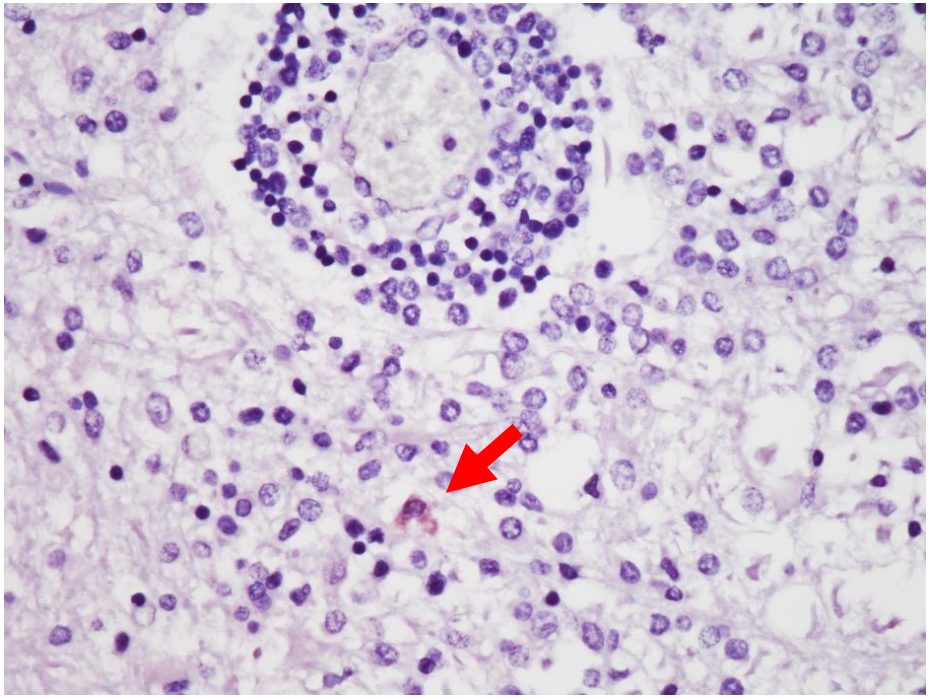


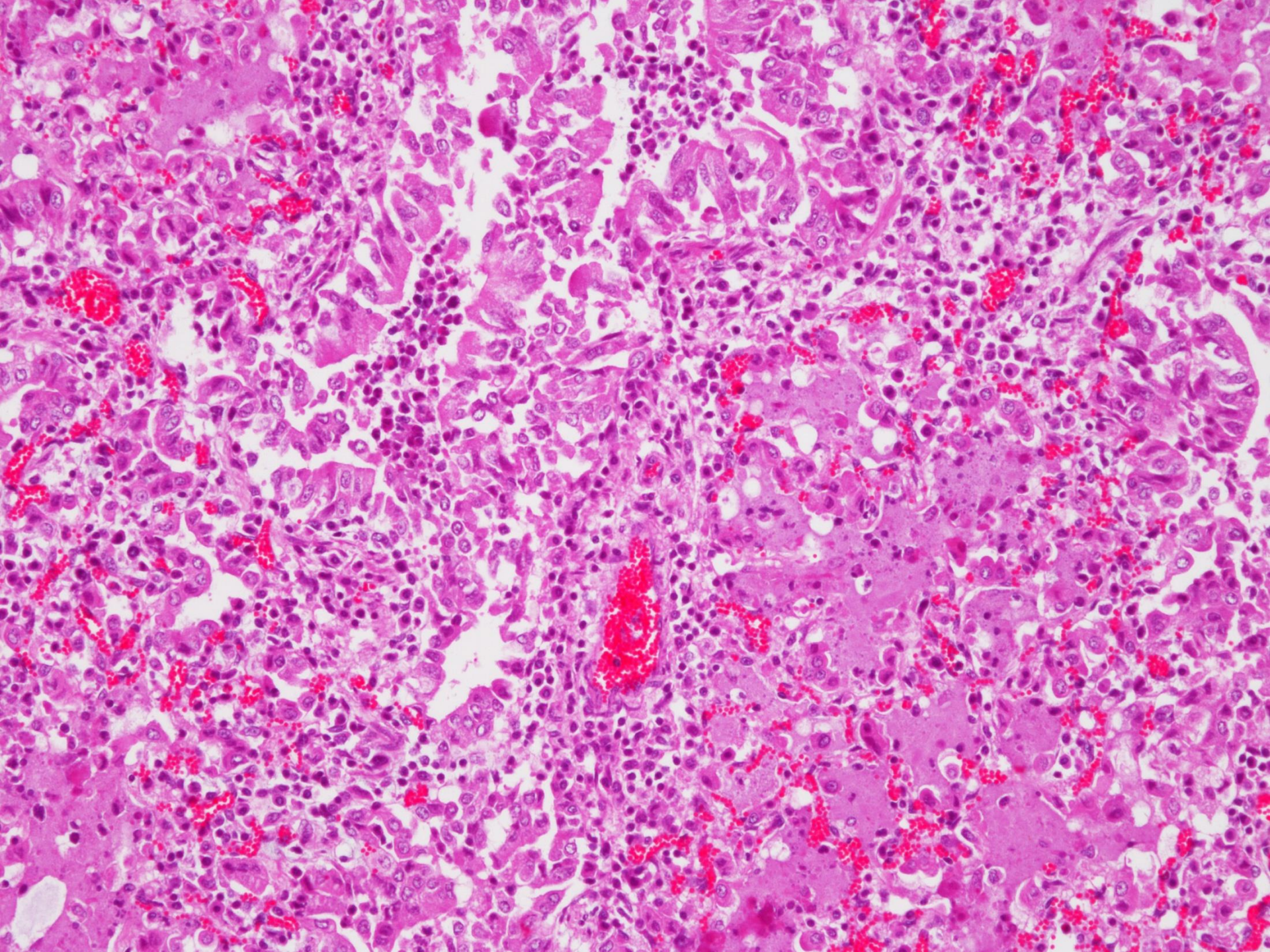


In situ Hibridization

- Help to co-localize the viral nucleic acid→ **Macrophage-tropism**







Prevention & Control

- **Segregating or culling** of seropositive animals
- Preventing/minimizing perinatal transmission of infection to the kid, particularly colostrum and milk transmission
- No effective vaccine available
- Marker-assisted genetic selection, to identify animals less susceptible to infection

Q Fever

Q Fever

- **Zoonosis / WOAH notifiable disease**
- **第四類法定傳染病 (2007)**
- In 1935, a typical pneumonia occurred in abattoir workers in Queensland, Australia, and was called "abattoir fever". Because the cause was a "query," it was named Q fever in 1937

第四類法定傳染病

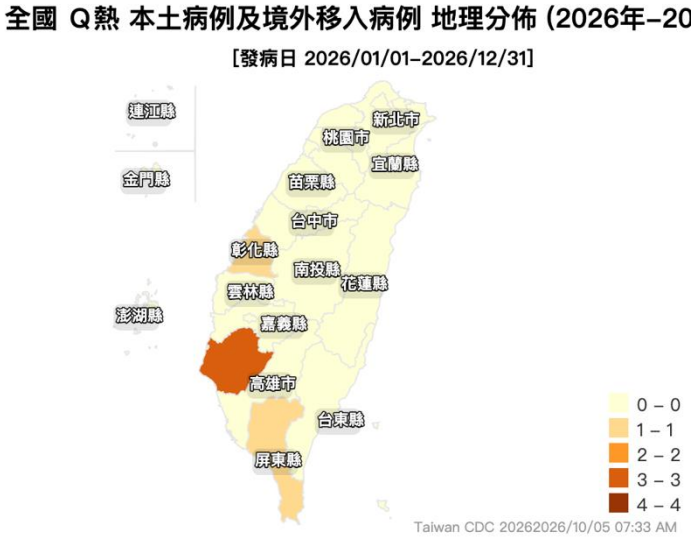
新冠併發重症	李斯特菌症	水痘併發症
恙蟲病	地方性斑疹傷寒	發熱伴血小板減少綜合症
萊姆病	肉毒桿菌中毒	庫賈氏病
弓形蟲感染症	布氏桿菌病	流感併發重症
侵襲性肺炎鏈球菌感染症	Q熱	類鼻疽
鉤端螺旋體病	兔熱病	疱疹B病毒感染症

https://www.cdc.gov.tw/Disease/Index

趨勢圖 | 地理分佈 | 圖表 | 同期比較 | 境外移入 | 疾病小百科



https://nidss.cdc.gov.tw/nndss/disease?id=0830



Etiology

- ***Coxiella burnetii,*** a **gram-negative**, **obligate intracellular** bacterium belonging to the class Gammaproteobacteria.
- Two morphologically distinct forms:
  -- Large cell variant: pleomorphic, metabolically active (replicating) and sensitive to environmental conditions
  -- **Small cell variant**: short-rod-shaped, nonreplicating, **resistant** to environmental stressors

對大多數的消毒劑都有抗性; 耐熱耐乾燥

- Two antigenic types:
  -- Type I: pathogenic in animals and the wild.
  -- Type II is produced by serial passages in chick embryo yolk sacs and cell cultures; less pathogenic.

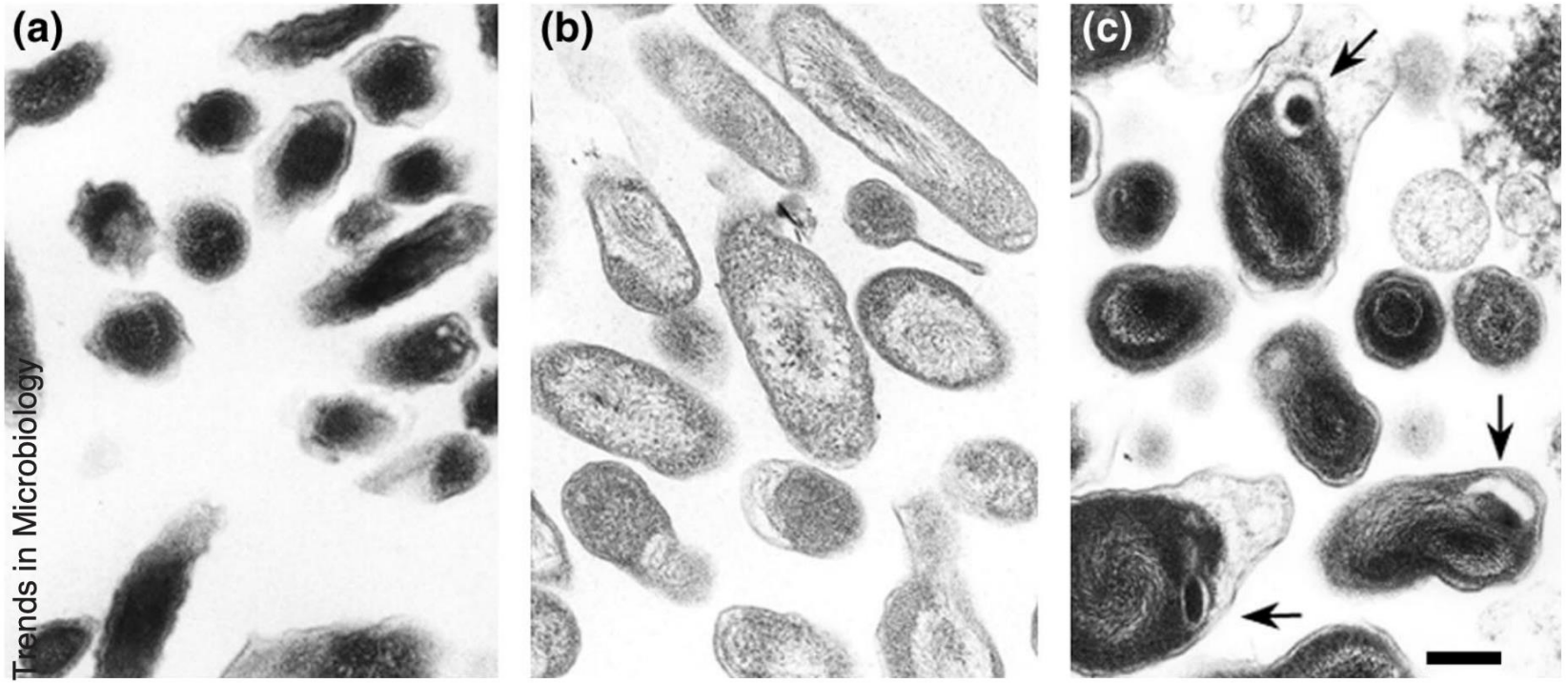


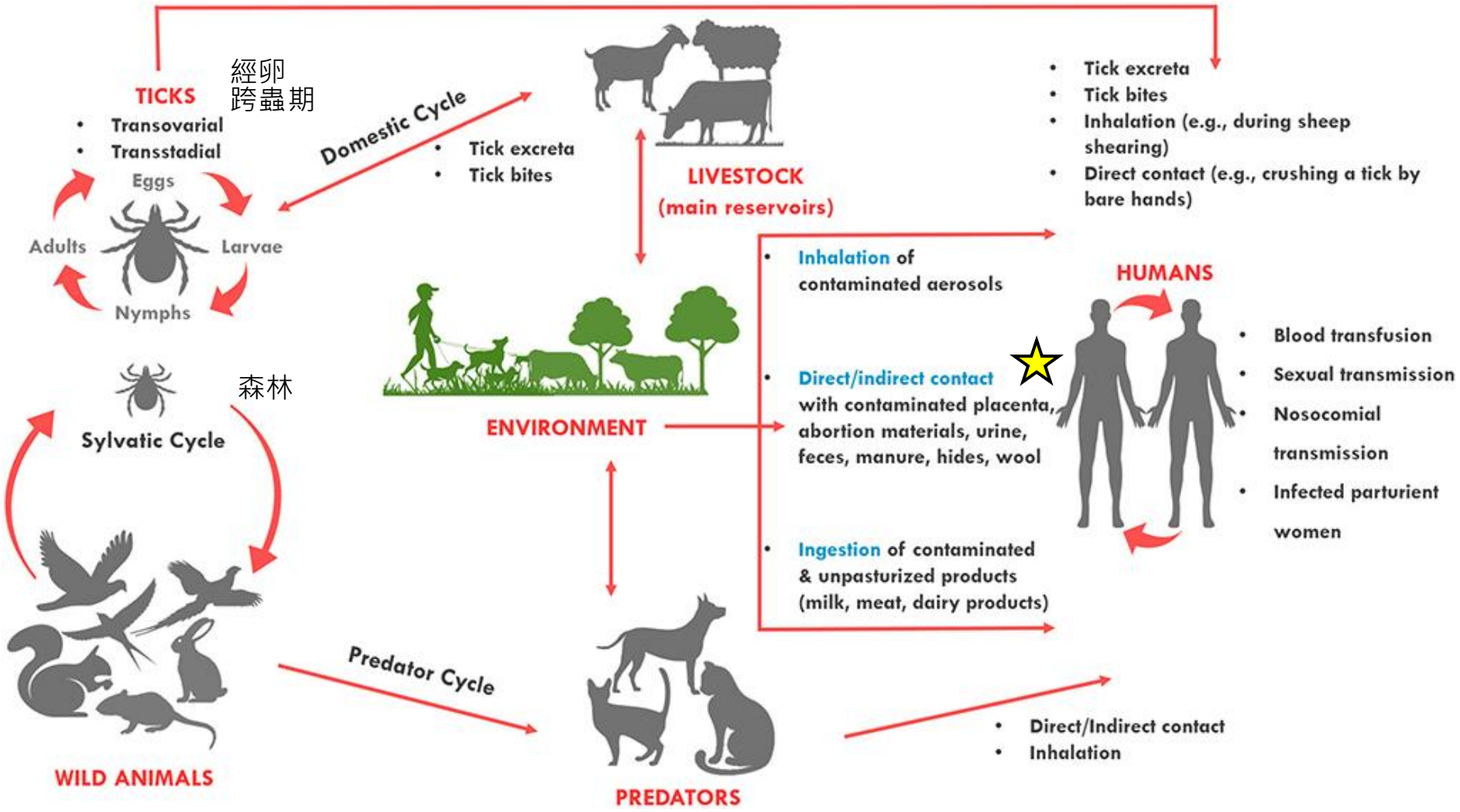
Fig. 2. Electron micrographs of *Coxiella burnetii* small cell variants (SCVs) and large cell variants (LCVs), and LCVs containing spore-like particles (SLPs). **(a)** Electron micrograph of purified SCVs. Note the size and characteristic condensed chromatin. **(b)** Electron micrograph of purified LCVs. Note the size and characteristic dispersed chromatin. **(c)** Electron micrograph of purified LCVs harboring SLPs (arrows). Scale bar = 0.2 μm.

Heinzen et al., 1999

Transmission

- **Worldwide distribution**
- **Inhalation of aerosols or dust containing the pathogen is the main route of transmission**. Other routes include skin wounds, conjunctival contact, and the digestive tract.
- The pathogen is present in the animal's mammary glands, mammary lymph nodes, uterus, and placenta.
- It is transmitted through **blood**, **milk**, feces, urine, amniotic fluid, and secretions.
- **Ticks** can also be a vector (rare).

Wide range of host species



Celina, S. S., & Cerný, J., 2022

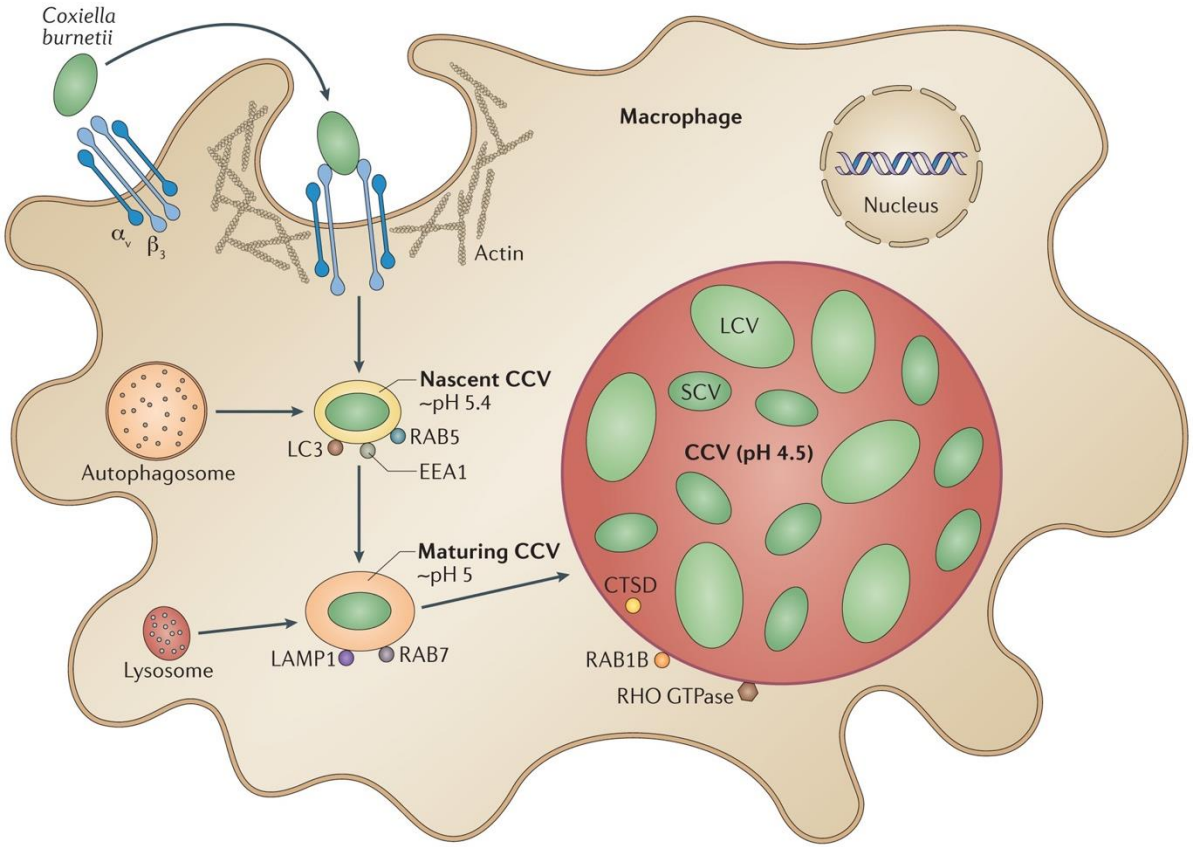
Pathogenesis

- It infects **monocytes and macrophages** and survives within the acidic environment (cue for replication) of the phagosome.

- High affinity for the animal's **trophoblasts in placenta**

10^9/g placenta

- Shed through uterine, placenta, feces, and milk
- After human inhalation, the pathogen first proliferates in the **lungs** and then circulates through the bloodstream to all organs.



Nature Reviews | Microbiology

Clinical Presentation

- Incubation period: varies greatly, from about 9 to 28 days.
- **Most ruminants are subclinically infected**. Possible symptoms include mild fever and various **reproductive disorders** including infertility, **abortion (孕後期),** stillbirths, weak offspring, retained placenta, and endometritis.
- In **humans:** the most common symptoms are **fever**, fatigue, headache, altered consciousness, nausea, vomiting, and diarrhea. It can lead to fatal "acute respiratory distress syndrome" due to atypical pneumonia. Other conditions include granulomatous hepatitis, hepatomegaly, and jaundice.

Chronic infection can also cause meningitis, orchitis, myocarditis, **endocarditis**, and nephritis

前置因子如心臟瓣膜疾病、血管缺損

Pathological Findings

- **Gross Pathology**: In animals, lesions are limited to the **placenta**, which is thickened, leathery, and has multifocal to coalescing areas of calcification.
- **Microscopic lesions**: In animals, **necrotizing intercotyledonary placentitis** with intracellular organisms +/- vasculitis. In humans, there are granulomatous nodules in various affected tissues

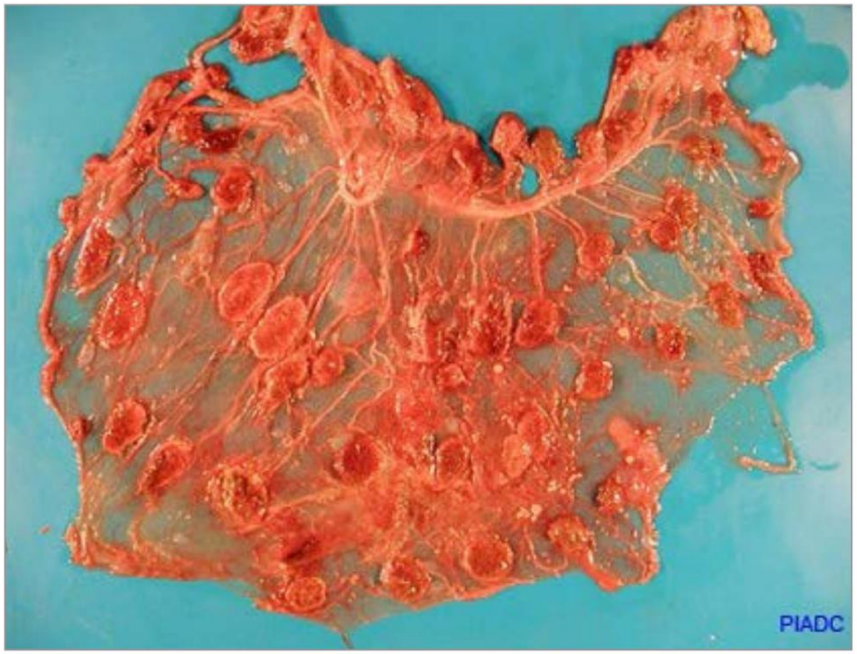
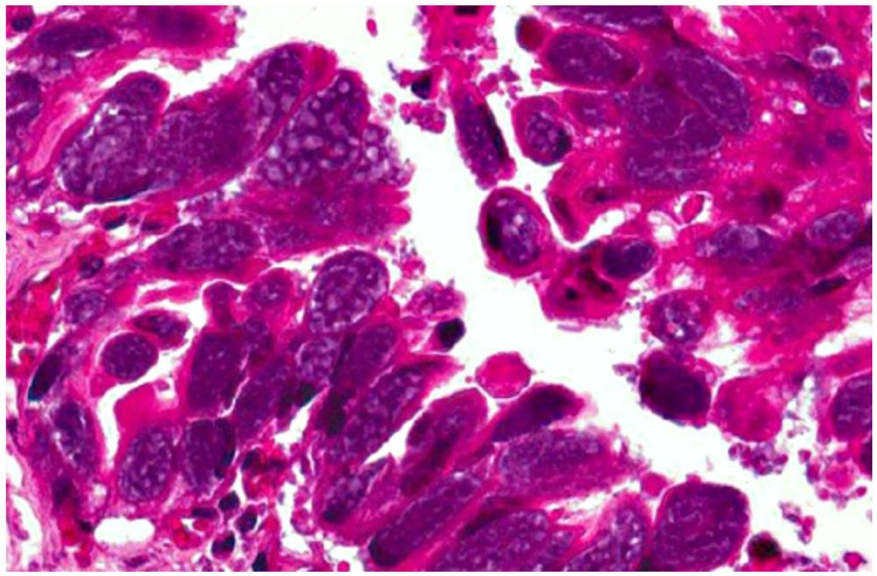


Figure 1. Placenta infected with *Coxiella burnetti.* Notice the thickened tan areas of the placenta.

https://www.extension.purdue.edu/extmedia/va/VA-22-W.pdf



2–3. Placenta (Hematoxylin and Eosin | –400x)
Higher magnification of chorionic epithelium demonstrated the characteristic vacuolated appearance of the intracytoplasmic bacterial inclusions of *Coxiella burnetti*

https://www.askjpc.org/wsco/do_slideshow2.php?pic_series=dEFnNVNVeEV5NndiMUVvVjNINk01UT09&cYear=2014&initImage=eDdPeEp4aGVCVmRqSmdYVVg5Yzk5Zz09

Diagnosis

- **P3** biosafety protection equipment and measures.
- **General Laboratory Diagnosis**:
  -- Find the organism: immunostaining
  -- Nucleic acid detection: PCR
  -- Serology to check the Ab response: Indirect immunofluorescence assay, ELISA, and CF.

Q 熱
（Q fever）

一、臨床條件

發燒、畏寒、噁心嘔吐、腹瀉、盜汗、頭痛、身體不適、肌肉酸痛，急性肝炎或急性黃疸、肺炎以及腦膜腦炎等。

二、檢驗條件

符合下列檢驗結果之任一項者，定義為檢驗結果陽性：

（一）臨床檢體分子生物學核酸檢測陽性。

（二）以間接免疫螢光抗體法（Indirect Immunofluorescene Assay，IFA）檢測急性期（或初次採檢）血清，Phase II IgM 力價達 1：80 以上，或配對血清（急性期及恢復期）Phase II IgG 力價達 4 倍以上上升。

Prevention & Control

Vaccination, antibiotic therapy, hygiene, and biosecurity

- Vaccination: reduce the incidence of disease, bacterial shedding, and prevent abortion, but it does **not prevent infection**
- Antibiotics**:** Chlortetracycline, Tetracycline, or Chloramphenicol.
  -- Lack of positive effects of using antibiotics to control clinical signs in animals as a single control measure
- Hygiene & biosecurity: **manure** removal, management of **placentas**, use of sanitized **protective clothes** by farm workers, and restriction of visitor access…

Enzootic Abortion of Ewes (EAE)

Enzootic Abortion of Ewes (EAE)

- **Zoonosis / WOAH notifiable disease**
- Similar disease presentation in goats
- Also cause abortion in **cattle**, **pigs**, and **horses** but the incidence is quite **low**.

Etiology

- ***Chlamydophila abortus***, is a small**, gram-negative**, obligate intracytoplasmic bacterium.
- Two forms:

  -- **Elementary body (EB)**: a small, dense sphere that can remain infectious for several months in **low-temperature** environments.

  -- Reticulate body (RB: non-infectious

Up to several months if the temperature is low or close to 0 ˚C.

- Resistant to acids, alkalis, toluene-based compounds (甲苯基化合物), and lime (石灰), but **sensitive to most disinfectants**, such as 1% sodium hypochlorite, 70% ethanol, glutaraldehyde, and formaldehyde; also inactivated by **heat**.

Transmission

- Introduction of latently infected ewe
- The main source of transmission is the **aborted placentas or uterine secretions** from sick ewes.
- The placenta and **uterine/vaginal secretions** of sick ewes contain a large amount of *C. abortus*.
- Infected ewes shed the organism for a week before aborting and 2 weeks afterward.
- Common routes include **oral or nasal inoculation**, **vertical transmission**, and the presence of *C. abortus* in the feces and urine of cattle and sheep, as well as in **goat milk.**

Epidemiology

- Worldwide distributed except for Australia and New Zealand
- **In Taiwan**: the seropositive rate of goats collected from six farms during 2007–2011 was 60% (18/30). Aborted fetuses checked for the presence of six abortigenic pathogens by PCR revealed highest positive rate for *Cp. abortus* of 26.1% (6/23). → **Major causative agent** for goat abortion.
- The main hosts are sheep and goats, but it can also infect cattle, deer, llamas, rabbits, and horses.
- In a newly infected flock, **over 30%** of pregnant ewes abort in the last 3 weeks of gestation or give premature birth to weak or dead lambs.

Pathogenesis

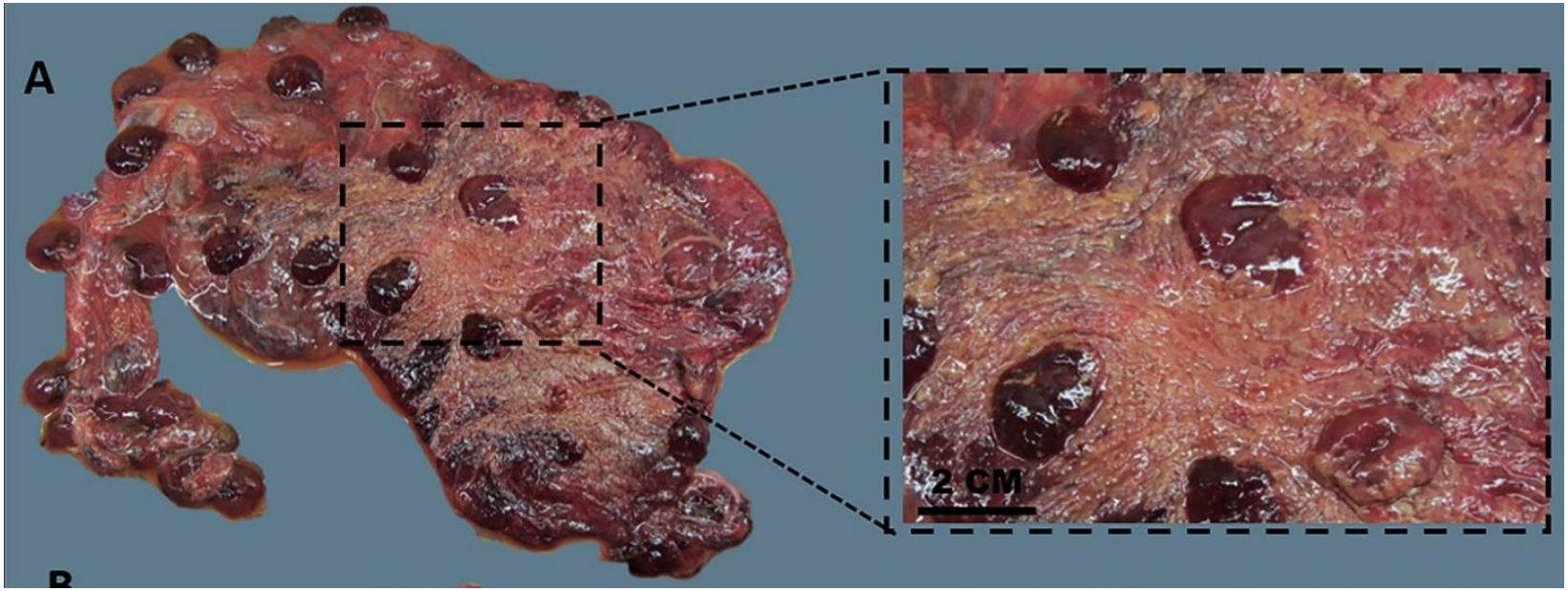
- Following infection, it is thought that the organism resides first in the **tonsil** and is then **disseminated by blood and lymph** to other organs, although the site of latent infection is not definitely known.
- Release from the **latent state** during pregnancy is thought to be caused by **immune modulation** and leads to bacteremia and infection of the placenta.

Clinical Presentation

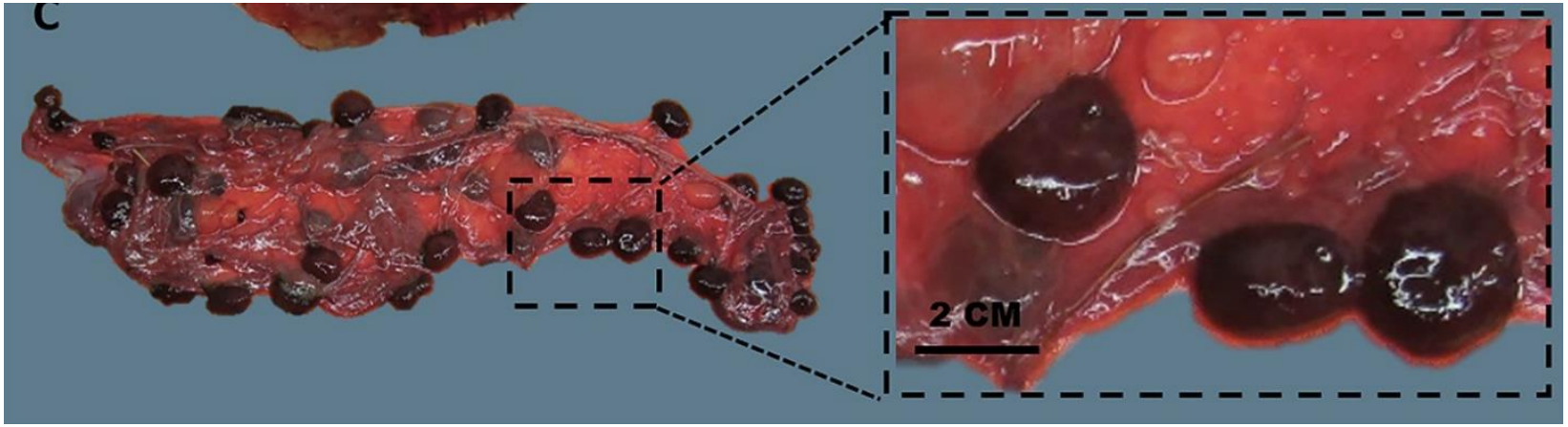
- Abortion, stillbirths, or the birth of weak lambs in **the last 2-3 weeks of pregnancy**.
- Pregnant ewes often **lack overt clinical signs**.
- In some cases, a small amount of reddish or brownish vaginal discharge may be seen before the abortion. Retained placenta and mastitis are not common.
- After the abortion, the ewe may recover quickly, but some may continue to have a **dark brown vaginal discharge**.

Pathological Findings

- Aborted fetuses:
  -- Typically no gross abnormalities
  -- Histologically, there may be mononuclear cell infiltration of hepatic portal areas and areas of **hepatitis**.
- Placental cotyledons are **necrotic and hemorrhagic**, and the intercotyledonary areas are thickened, edematous, and leathery. **Vasculitis** are more common than infection with *C. burnetiid.*

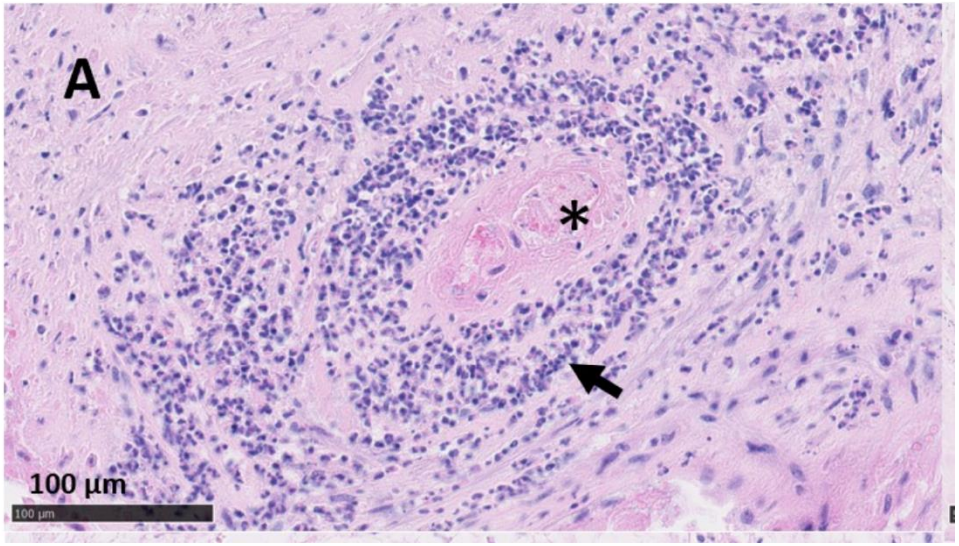


Oedema and thickening of the whole placenta covering and the typical dark red or grey-whitish colouration of the cotyledons and partially covered with a cream-coloured exudate on the intercotyledonary areas.

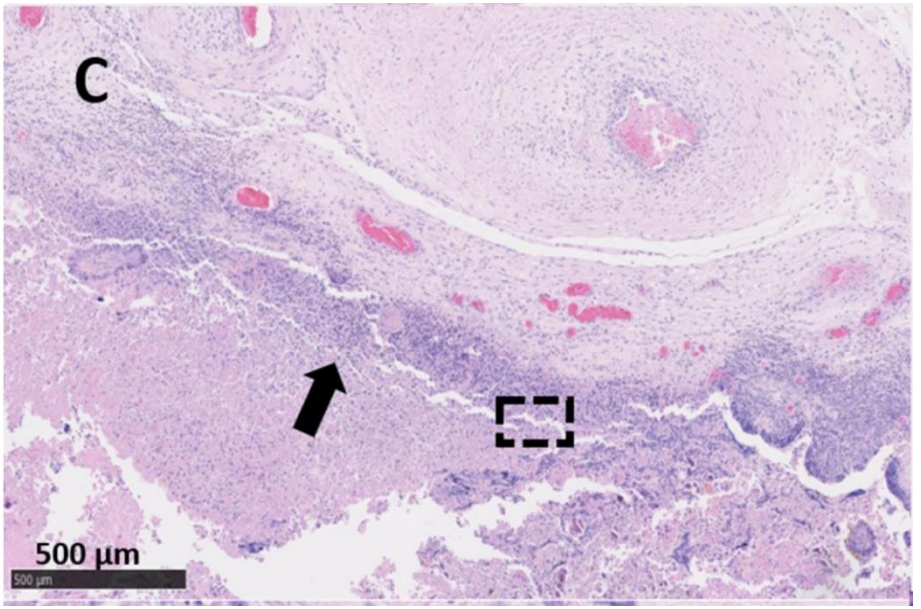


Normal placenta: red colour of the cotyledons, and thin and clear and translucent intercotyledonary membranes

Caspe et al., 2020



Necrosuppurative vasculitis, with severe mixed inflammatory infiltration (black arrow) and partially occlusive thrombosis (*);



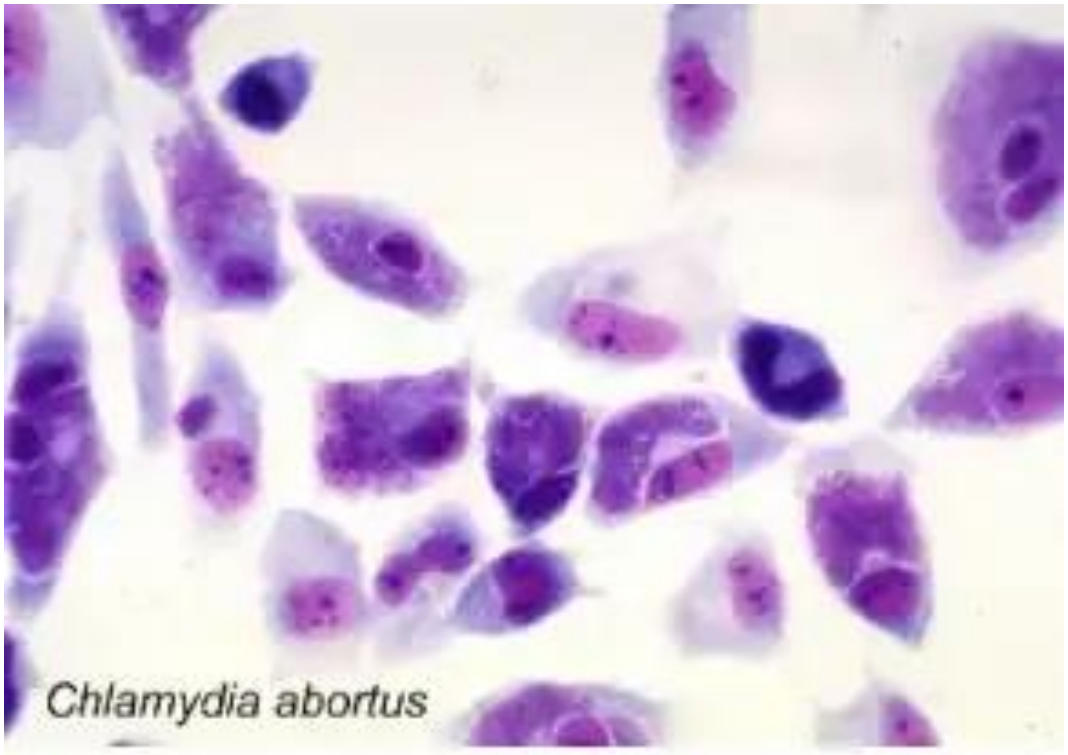
Necrosuppurative placentitis forming a compact layer consisting primarily of large numbers of leucocytes in the epithelium of the cotyledon (arrow) attached to superficial amorphous necrosuppurative material.

Caspe et al., 2020

Diagnosis

C. abortus should be considered when ewes have abortions, stillbirths, or **weak lambs in late pregnancy**.

- Laboratory Diagnosis:
  -- Tissue smear : smear from the sick ewe's placenta and uterine secretions and examine for *C. abortus* EBs after Giemsa or acid-fast stains.
  -- *C. abortus* EBs cannot be differentiated from *C. burnetii*.
- ELISA, IFA, PCR or pathogen isolation



https://www.vetmed.auburn.edu/academic-departments/dept-of-pathobiology/diagnostic-services/molecular-diagnostics/chlamydia-spp/

Prevention & Control

- Separating C. abortus-infected ewes from their lambs and treating them with long-acting oxytetracycline is the main way to control the persistent infection in the flock.
- Vaccination (either live or inactivated works) can reduce abortions in ewes but cannot prevent *C. abortus* infection in lambs.
- The zoonotic potential warrants caution when handling potentially infected material, and specimens should be properly packaged when submitted.